MODELS TO MEDICINES

Welcome: Logistics and Agenda

ACoP 2026

Accelerating Drug Development with an Innovative Clinical Utility Index (CUI) Approach

MODELS TO MEDICINES

Workshop Instructors:
Jesmin LohyDas (AstraZeneca)
Venkatesh Pilla Reddy (Eli Lilly)
Samer Mouksassi (Certara)

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Accelerating Drug Development with an Innovative Clinical Utility Index (CUI) Approach

USA



Jesmin LohyDas, PhD
AstraZeneca
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Samer MOUKSASSI, PhD
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Venkatesh Pilla Reddy, PhD
Eli Lily Co.
Arlington Square Site, UK

15th Oct 2026

ACoP 2026

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8:30 AM - 8:35 AM Welcome and Logistics:
CUI and other forms of multi-attribute utility approaches
Speaker: Samer Mouksassi, PharmD, PhD, FCP, FISoP – UCSF / Certara

8:35 AM - 9:00 AM EDT
What is CUI / Concept/Why is it important/Evolution of CUI in drug development
Speaker: Jesmin Lohy Das, MPharm, PhD – AstraZeneca

9:00 AM - 9:45 AM EDT
CUI for quantitative benefit – risk assessment in Drug Development Decision – Oncology
Speaker: Venkatesh Pilla Reddy, PhD (he/him/his) – Eli Lilly and Company Limited

10:00 AM - 10:30 AM EDT
CUI for quantitative benefit – risk assessment in Drug Development Decision – Non-Oncology
Speaker: Samer Mouksassi, PharmD, PhD, FCP, FISoP – Certara

10:30 AM - 10:45 AM EDT Tutorial Coffee Break
Location: Foyer

10:45 AM- 12:00 AM EDT - Demo / Hand-On
1. ADC (Oncology)
2. Elagolix (Non Oncology)

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CUI and other forms of multi-attribute utility approaches

Samer MOUKSASSI, PharmD, PhD, FCP, FISoP

Senior Director – Application Scientist
Accelerated Clinical Evidence Certara Radnor PA

ACoP 2026 Workshop; 15th October 2026

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The MAU family tree — CUI is one branch

CUI

Lalonde / Poland / Ouellet (CPT, 2009)

Expert-constructed utility functions, combined additively. Dose selection & benefit–risk.

Desirability functions

Derringer & Suich (1980)

The pre-CUI industrial-statistics cousin. Still the default in CMC / formulation optimization.

AHP

Saaty (1980)

Analytic Hierarchy Process, a pairwise-comparison weight elicitation. More common in portfolio & decision-board settings.

MCDA / PrOACT-URL

EMA & FDA benefit–risk frameworks

The regulator-facing sibling — structured, auditable benefit–risk assessment.

Conjoint analysis / DCE

"group CUI" variant

Weights elicited directly from patients or clinicians, not expert-constructed.

PTRS-weighted utility

Portfolio decision analysis

Folds probability of technical & regulatory success into the expected-utility calculation.

Health economics applies the same expected-utility logic to reimbursement decisions — out of scope here.

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Decision theory's deep roots

1944

Von Neumann & Morgenstern

Games and Economic Behavior gives utility its first axiomatic foundation — rational choice under uncertainty.

1961

Ward Edwards

Brings Bayesian methods into psychology; works out additive utility functions in detail.

1976

Keeney & Raiffa

Multi-Attribute Utility Theory (MAUT) — the shared mathematical ancestor of every method on this deck.

The throughline: *by 1976 the theory for combining several competing attributes into one defensible number already existed.*

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The 40-year gap

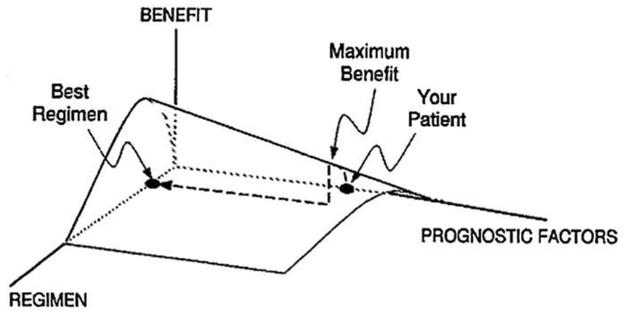
Theory existed

- Francis Anscombe (1963, 1990) flags the problems with Neyman–Pearson testing in clinical trials.
- Proposes a Bayesian “decision function” — net profitability from economic valuation.
- No named utility function yet, but the logic is already there.

Industry wasn't ready

- The approach lies essentially dormant, largely ignored by pharma.
- Berry and colleagues (1995, 2006) revive it — Bayesian sequential designs with financial evaluation.
- Still missing: quantitative exposure–response models to feed the utility function.

What finally closed the gap: ***model-based drug development*** *gave the fieldreal* ***dose–response surfaces*** *to put inside the* ***utility function****.*



Sheiner LB. Learning versus confirming in clinical drug development. Clin Pharmacol Ther [Internet]. 1997;61(3):275–91

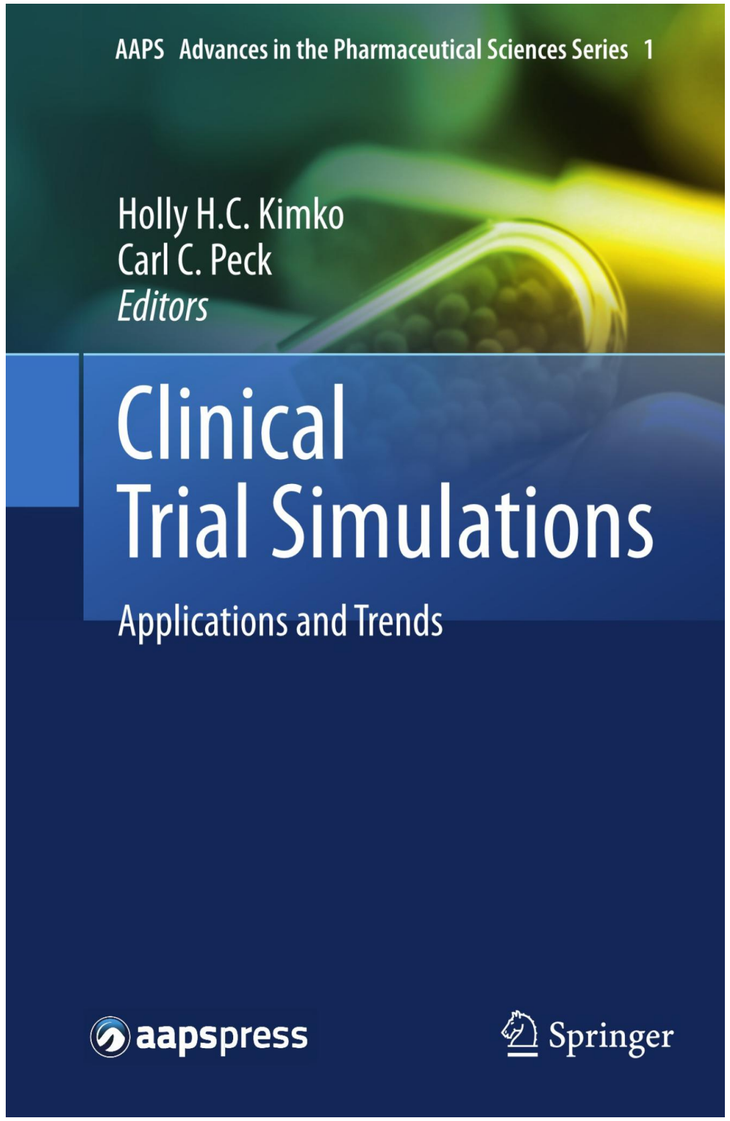
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Chapter 5
Decision-Making in Drug Development: Application of a Clinical Utility Index[SM]

Timothy J. Carrothers, F. Lee Hodge, Robert J. Korsan, William B. Poland, and Kevin H. Dykstra

H.H.C. Kimko and C.C. Peck (eds.), *Clinical Trial Simulations*, AAPS Advances in the Pharmaceutical Sciences Series 1, DOI 10.1007/978-1-4419-7415-0_5,
© American Association of Pharmaceutical Scientists 2011

Korsan B, Dykstra K, Pullman W (2005) Transparent tradeoffs: a clinical utility index openly evaluates a product's attributes and chance of success. Pharmaceutical Executive, March 2005

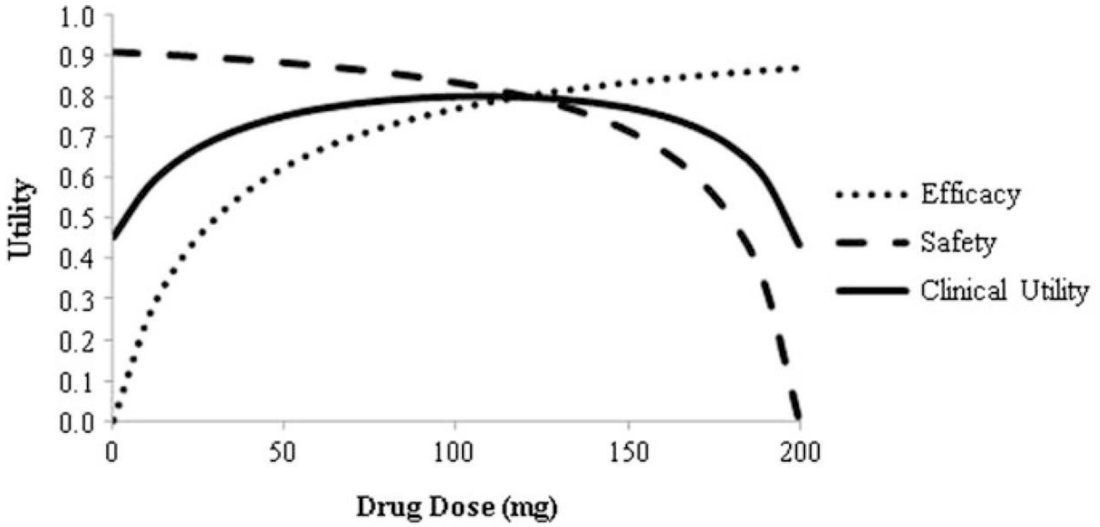


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Preview: the CUI mechanism (Carrothers et al., 2011)

Chapter 5
Decision-Making in Drug Development: Application of a Clinical Utility Index^{SM}

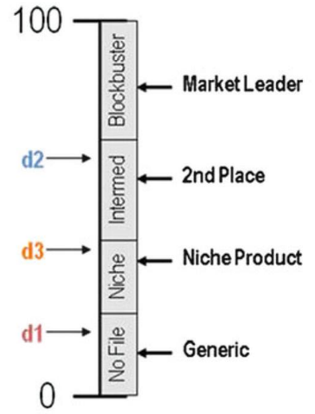
Timothy J. Carrothers, F. Lee Hodge, Robert J. Korsan, William B. Poland, and Kevin H. Dykstra



Note: Here, 'Safety' refers to relative lack of drug-related side effects. "Negative" attributes will decrease in utility with greater incidence (or value); thus, higher levels of utility correspond to lower levels of the adverse event.

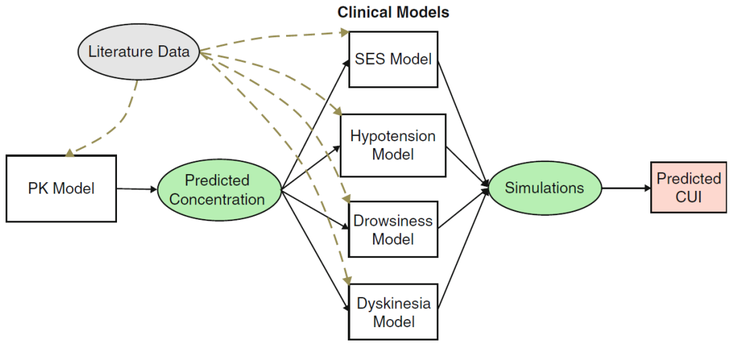
Fig. 5.1 A basic illustration of the utility concept

Fig. 5.2 Using utility to link to commercial viability



Note: d1, d2, and d3 are possible product profiles. "No File" (i.e., "no-go"), "Niche", "Intermed.", and "Blockbuster" represent discretization of the 0-100 utility range into market-meaningful categories. These categories are defined based on the translation of the known profiles of currently marketed products ("Market Leader", "2nd Place", "Niche Product", and "Generic") with specific levels of known commercial success (or lack thereof).

98 T.J. Carrothers et al.



Note: Figure depicts the steps in an integrated CUI-pharmacometric modeling project, with detail on the pharmacometric steps. Literature data is used to guide the construction of the PK and exposure-response models, which are based on subject-level data where possible. "Simulations" incorporates the integration of the CUI model with the PM output, as previously depicted in Figure 5.3 Predicted CUI is the variable of final interest.

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Modern understanding: what's moved since 2009

1. **The independence assumption is shaky —** efficacy and safety attributes are often correlated through the same underlying exposure, not truly independent inputs.

2. **Weight elicitation is still the soft underbelly —** clinician, patient, regulator, and sponsor weights diverge; FDA's Patient-Focused Drug Development program now feeds real preference data directly in.

3. **Sensitivity / robustness analysis is now expected —** tornado diagrams on the weights, not a single point score presented as fact.

4. **Bayesian / probabilistic utility —** treating the utility itself as a distribution, propagating PK/PD parameter uncertainty through to the decision.

5. **No FDA / EMA mandate for any single method —** unlike the near-consensus on meta-analysis forest-plot conventions, this space remains genuinely unsettled.

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The honest critique

False precision

A single utility score can hide exactly where the real disagreement lives — two stakeholders can agree on the number and disagree sharply on why.

Gameable by design

Weights chosen after seeing the data quietly launder a preferred outcome into an “objective” score. Pre-specification matters as much as the math.

The fix isn't abandoning MAU — *it's transparency: pre-specified weights, published sensitivity analyses, and multiple stakeholders setting the weights together.*

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What is CUI / Concept/Why is it important/Evolution of CUI in drug development

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What is CUI / Concept/Why is it important/Evolution of CUI in drug development

Jesmin Lohy Das, MPharm PhD

Astra Zeneca Boston, MA, USA

ACoP 2026 Workshop; 15th October 2026

AstraZeneca

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- The presenter has no relevant financial relationships to disclose in connection with this presentation.

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Who’s in the Room?

https://www.mentimeter.com/app/presentation/alhsibn8qeawmojz2h3h353ine2mqkx3/edit?question=doeqecymyub4

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2007–2009: the CUI is born

2007

Lalonde et al. — Model-Based Drug Development[1]

CPT paper codifies the quantitative-pharmacology paradigm; gives the field exposure–response models to act on.

2009

Ouellet, Lalonde et al. — first applied CUI[2]

Two insomnia compounds scored on sedation + 5 efficacy measures; the backup compound's peak utility sat at non-viable doses — program discontinued.

2009

Poland, Hodge et al. — CUI formalized[3]

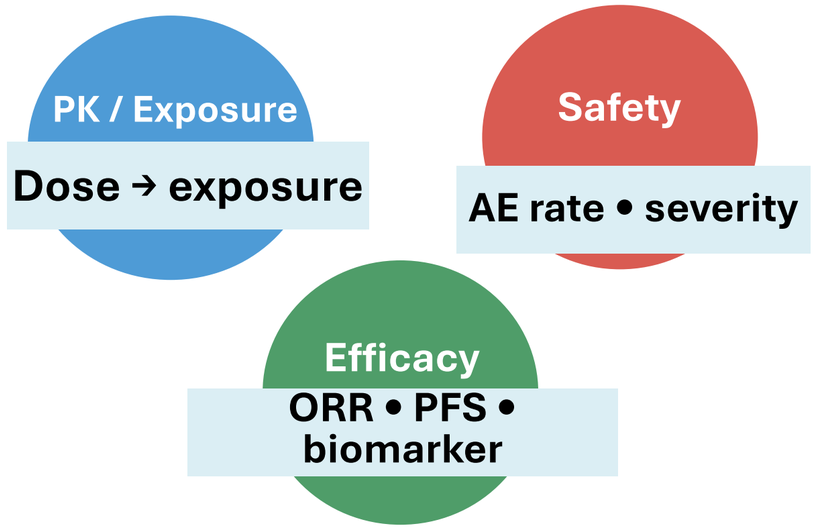
Companion CPT paper generalizes the method as a practical multiattribute (MAUT-based) framework for drug-development decisions.

1. Lalonde RL et al. Model-based drug development. Clin Pharmacol Ther. 2007;82(1):21–32.
2. Ouellet D, Werth J, Parekh N, Feltner D, McCarthy B, Lalonde RL. The use of a clinical utility index to compare insomnia compounds: a quantitative basis for benefit-risk assessment. Clin Pharmacol Ther. 2009;85(3):277–282.
3. Poland B, Hodge FL, Khan A, Clemen RT, Wagner JA, Dykstra K, Krishna R. The clinical utility index as a practical multiattribute approach to drug development decisions. Clin Pharmacol Ther. 2009;86(1):105–108.

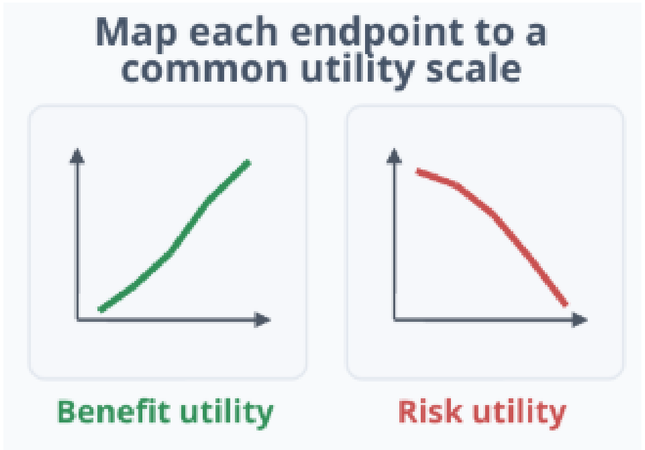
What is Clinical Utility Index (CUI) ?

Quantitative framework that integrates multiple measures of clinical benefit and risk into
One Single Comprehensive Measure for Optimum Biological Dose (OBD)

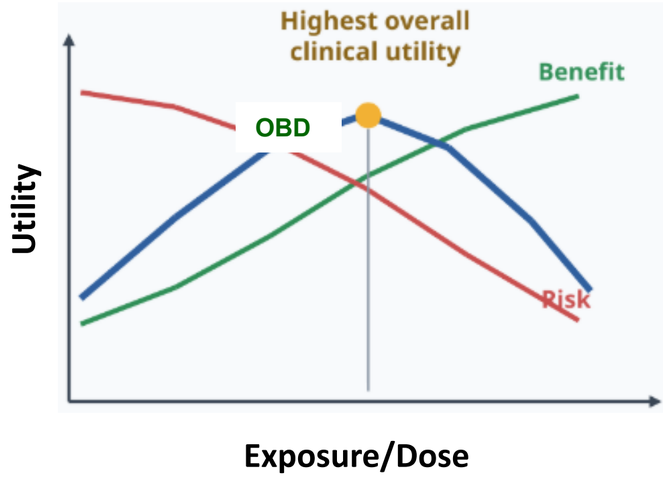
MULTIPLE OUTCOMES
(CLINICALLY RELEVANT EVIDENCE)



COMMON VALUE SCALE
(TRANSLATE INTO UTILITY)



DECISION SUPPORT
(INTEGRATE BENEFIT AND RISK)



"Which option provides the best overall clinical value?"
—not simply the highest efficacy or the lowest toxicity.

Why do we need CUI ?

Different dimensions may point to different ‘best’ choices

BENEFIT

How much clinical benefit?

- Response / symptom control
- Durability / disease control
- Meaningful change for patients

RISK

What is the clinical cost?

- Adverse events
- Treatment discontinuation
- Long-term tolerability

EXPOSURE / PK–PD

What drives the outcomes?

- Dose and exposure
- Variability between patients
- Exposure–response relationships

Decision question:

Which option provides the best overall clinical value — not simply the best individual endpoint?

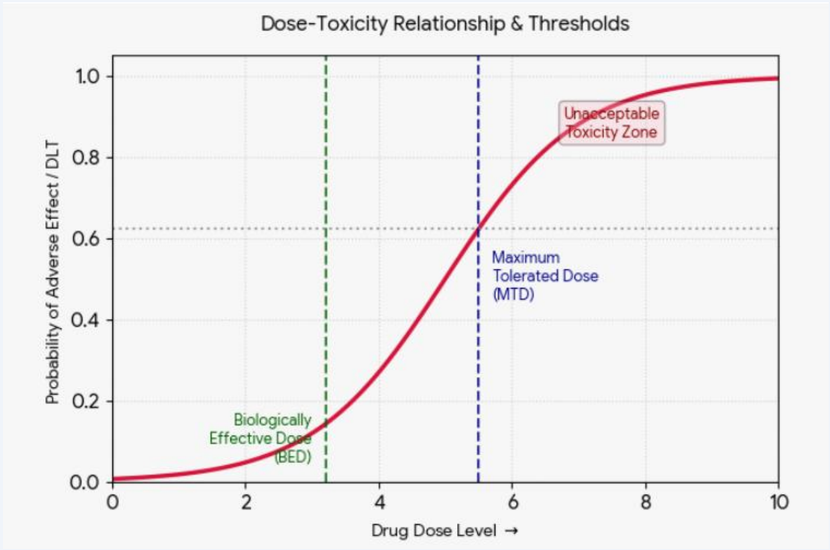
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Why do we need CUI ?

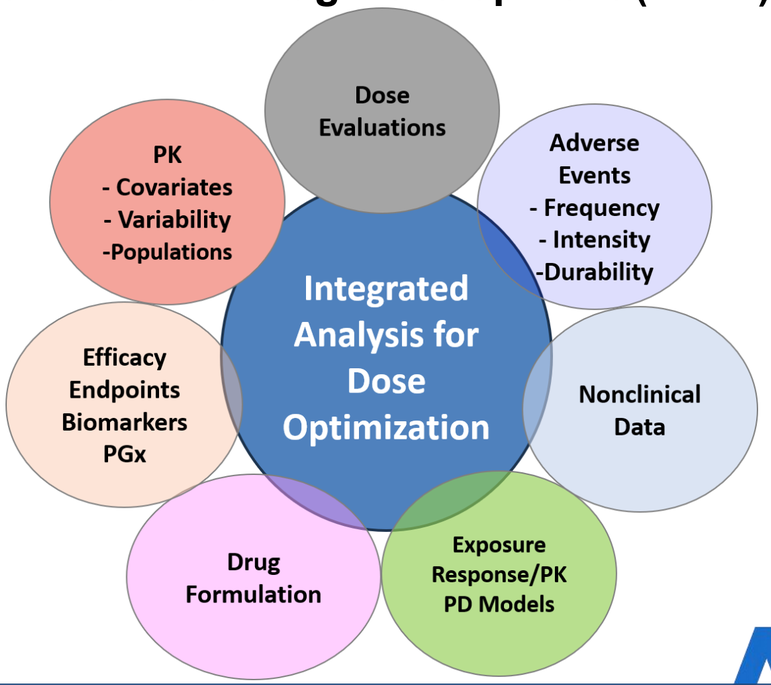
PROJECT OPTIMUS

TRADITIONAL PARADIGM → **THE SHIFT** → **OPTIMIZATION PARADIGM**

Maximum Tolerable Dose (MTD)



Model Informed Drug Development (MIDD)



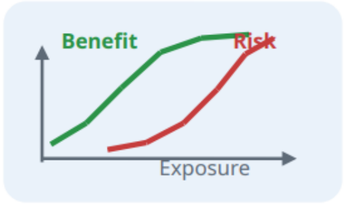
Clinical Utility Index is one of the tools in the MIDD toolbox to implement the Vision of Project Optimus

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How is CUI Implemented?

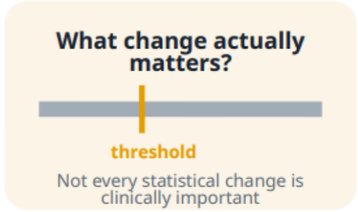
Four elements turn multiple efficacy and safety signals into a transparent, testable decision framework

1 Characterize Exposure–Response



- Efficacy endpoints
- Safety / tolerability
- Across dose or exposure

2 Define Clinical Meaning



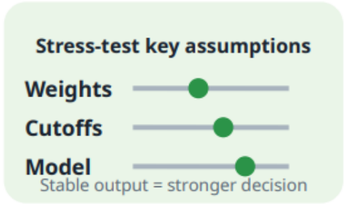
- Meaningful benefit
- Acceptable / unacceptable risk
- Clinically relevant thresholds

3 Select & Weight Attributes

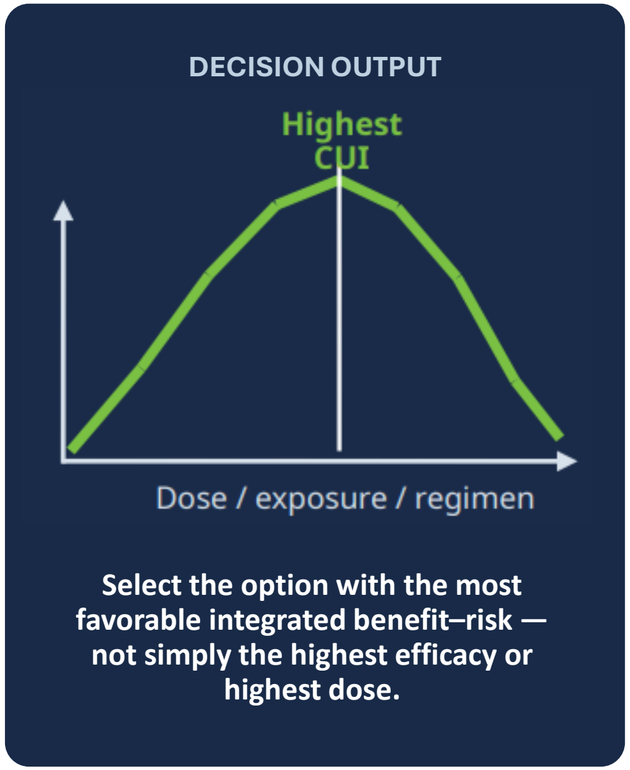


- Choose decision-driving outcomes
- Translate to comparable utility
- Set relative importance

4 Test Robustness



- Sensitivity to weights / cutoffs
- Parameter uncertainty
- Scenario analysis



Key principle:
Clinical meaning and uncertainty are defined explicitly before the CUI is used to support a decision.

Ouellet D. Benefit–risk assessment: the use of clinical utility index. Expert Opin Drug Saf. 2010;9(2):289–300. doi:10.1517/14740330903499265

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Where Can CUI Support Drug Development?

CUI is a decision framework — not only a dose-selection tool

The same benefit–risk logic can be applied at multiple decision points across development.

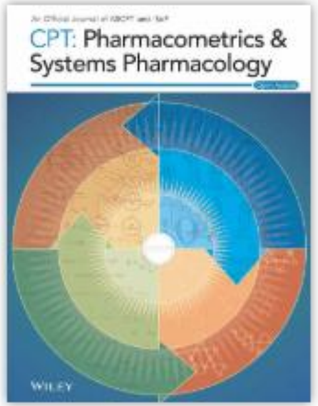
1	2	3	4	5
Discovery / Preclinical	**Phase 1**	**Phase 2**	**Phase 3 / Registration**	**Post-Approval / Lifecycle**
				
Candidate or regimen comparison	**First-in-human dose exploration**	**Dose optimization**	**Confirmatory strategy**	**Formulation, population, or regimen optimization**
Go / no-go for proof of concept	Schedule / dose-range refinement	Dose or regimen selection	Quantitative benefit–risk support	Lifecycle decision support

The decision question comes first.
The CUI attributes, utility functions, and weights should be tailored to that decision context.

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PUBLICATIONS

Postmarketing Assessment of Antibody–Drug Conjugates: Proof-of-Concept Using Model-Based Meta-Analysis and a Clinical Utility Index Approach



Innocent Gerald Asiimwe, Nour Chtiba, Samer Mouksassi, Goonaseelan (Colin) Pillai, Raimund M. Peter, Eunice Yuen, Venkatesh Pilla Reddy

First published: 04 March 2025 | https://doi.org/10.1002/psp4.70013

CPT: Pharmacometrics & Syst Pharmacol (2025)

Clinical Pharmacology & Therapeutics

Articles | Full Access

Moving Beyond Maximum Tolerated Dose for Targeted Oncology Drugs: Use of Clinical Utility Index to Optimize Venetoclax Dosage in Multiple Myeloma Patients

KJ Freise, AK Jones, ME Verdugo, RM Menon, PC Maciag, AH Salem

First published: 17 April 2017 | https://doi.org/10.1002/cpt.712

The AAPS Journal, Vol. 11, No. 1, March 2009 (© 2009)
DOI: 10.1208/s12248-008-9074-z

Review Article

Theme: Quantitative Pharmacology, a Roadmap for Rational, Model-Based Drug Development
Guest Editor: Bernd Meibohm

The Use of Clinical Utility Assessments in Early Clinical Development

Anis A. Khan,[1] Itay Perlstein,[1] and Rajesh Krishna[1,2]

Received 27 August 2008; accepted 8 December 2008; published online 16 January 2009

Abstract. A quickly realizable benefit of model-based drug development is in reducing uncertainty in risk/benefit, comprising individually of safety and effectiveness, two key attributes of a product evaluated for regulatory approval, marketing, and use. In this review, we investigate gaps and opportunities in using fundamental decision analytic principles in drug development and present a quantitative clinical pharmacology framework for the application of such aids for early clinical development decision making. We anticipate that implementation of such emerging tools will enable sufficient scientific understanding of the two attributes to facilitate the early termination of compounds with less than desirable risk/benefit profiles and continuance of compounds with acceptable risk/benefit profiles.

KEY WORDS: clinical utility index; modeling and simulation; therapeutic index.

POSTER PRESENTATIONS - PROFFERED ABSTRACTS | APRIL 04 2023

Abstract 5668: Using clinical utility index (CUI) to determine the optimal biological dose of a nonfucosylated anti-TIGIT antibody: A proposed alternative to maximum tolerated dose (MTD)

Gabriela Patilea-Vrana; John Harrold; Joseph A. Ware; Shaparak Lonning; Hun Lee; Lisa Brooks; Haley Neff-LaFord; William D. Hanley; Andres Forero-Torres

Cancer Res (2023) 83 (7_Supplement): 5668.
https://doi.org/10.1158/1538-7445.AM2023-5668

A Clinical Utility Index for Selecting an Optimal Insulin Dosing Algorithm for LY2605541 in Patients with Type 2 Diabetes Pretreated with Basal Insulin

David H. Manner, PhD, Junxiang Luo, PhD, [...], and Scott J. Jacober, DO (+3) View all authors and affiliations

Volume 16, Issue 8 | https://doi.org/10.1089/dia.2013.0268

ATTD Advanced Technologies & Treatments for Diabetes

Citation: CPT Pharmacometrics Syst. Pharmacol. (2019) 8, 240–248; doi:10.1002/psp4.12394

ARTICLE

Exposure–Response-Based Product Profile–Driven Clinical Utility Index for Ipatasertib Dose Selection in Prostate Cancer

Rui Zhu[1,*], Bill Poland[2], Russ Wada[2], Qi Liu[1], Luna Musib[1], Daniel Maslyar[1], Eunpi Cho[1], Wei Yu[1], Han Ma[1], Jin Yan Jin[1] and Nageshwar Budha[1]

Citation: CPT Pharmacometrics Syst. Pharmacol. (2021) 10, 40–47; doi:10.1002/psp4.12570

ARTICLE

A Personalized Medicine Approach Using Clinical Utility Index and Exposure-Response Modeling Informed by Patient Preferences Data

Insa Winzenborg[1], Ahmed M. Soliman[2] and Mohamad Shebley[3,*]

Selection of a personalized dose for an individual patient can be informed by the patient's preferences, translated as weights on each of the clinically relevant safety and efficacy drug attributes, based on results from a brief patient preference elicitation questionnaire. In this analysis, the weighted attributes were simulated to represent various endometriosis patient pro-

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CUI for quantitative benefit – risk assessment in Drug Development Decision – Oncology

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Integrating Clinical Utility Index Modeling Approach for Antibody-Drug Conjugates

Venkatesh Pilla Reddy, PhD

Global PK/PD Pharmacometrics, Eli Lilly and Company, UK

ACoP 2026 Workshop; 15th October 2026

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- We have no financial relationships to disclose relating to this presentation

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Agenda

- Quantitative Medicine in the Context of Project Optimus for ADCs
- Clinical Utility Index (CUI) Approach
- Case Studies
- Demonstration of CUI Shiny App
- Conclusions

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Project Optimus

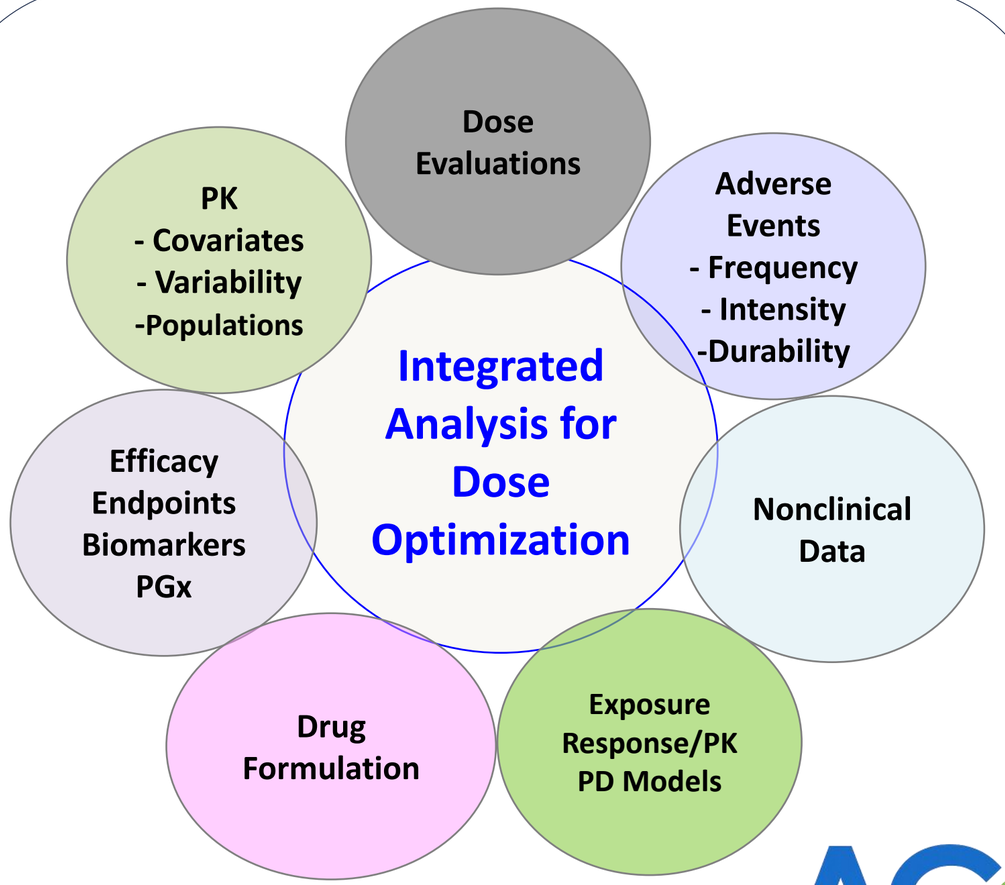
FDA's initiative to transform how we select the right dose for oncology drugs

Shift towards Dose-Optimization Paradigm in oncology drug development from Maximum Tolerable Dose

- ✓ Enhance characterization of doses in randomised settings prior to registration trials to maximize the efficacy and safety of oncology drugs
- ✓ Inadequate dose justification may not be acceptable

https://www.fda.gov/about-fda/oncology-center-excellence/project-optimus

Model Informed Drug Development (MIDD)



Clinical Utility Index is one of the tools in the MIDD toolbox to implement the Vision of Project Optimus

PK =Pharmacokinetics, PD= Pharmacodynamics

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FDA Dose Optimisation Guidance

GUIDANCE DOCUMENT

Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases

Guidance for Industry; Availability

III. DOSAGE OPTIMIZATION RECOMMENDATIONS

Dosages selected for administration in a clinical trial(s) should be adequately supported by data appropriate to the stage of development. Relevant nonclinical[6] and clinical data (such as PK, PD, safety, tolerability, dosage convenience, and activity), as well as the dose- and exposure-response relationships should be evaluated to select a dosage(s) for clinical trial(s). An approach where a dosage is chosen for a trial without adequate justification or consideration of all relevant data may not be acceptable, because FDA may determine that patients are exposed to unreasonable and significant risk, or there is insufficient information to determine risk, or the design of the trial is deficient to meet its stated objectives and may place a protocol on clinical hold.[7]

Sponsors, including those pursuing development of a drug under an FDA expedited program (e.g., breakthrough therapy designation), should plan their development programs such that identification of an optimized dosage(s) can occur prior to or concurrently with the establishment of the drug's safety and efficacy. Sponsors should note that development of a drug under an FDA expedited program is not a sufficient justification to avoid identifying an optimized dosage(s) prior to submitting a marketing application.

FDA ADC Guidance: Quantitative Modeling Is Now the Standard

WHAT FDA GUIDANCE RECOMMENDS

1 Dosing Strategies

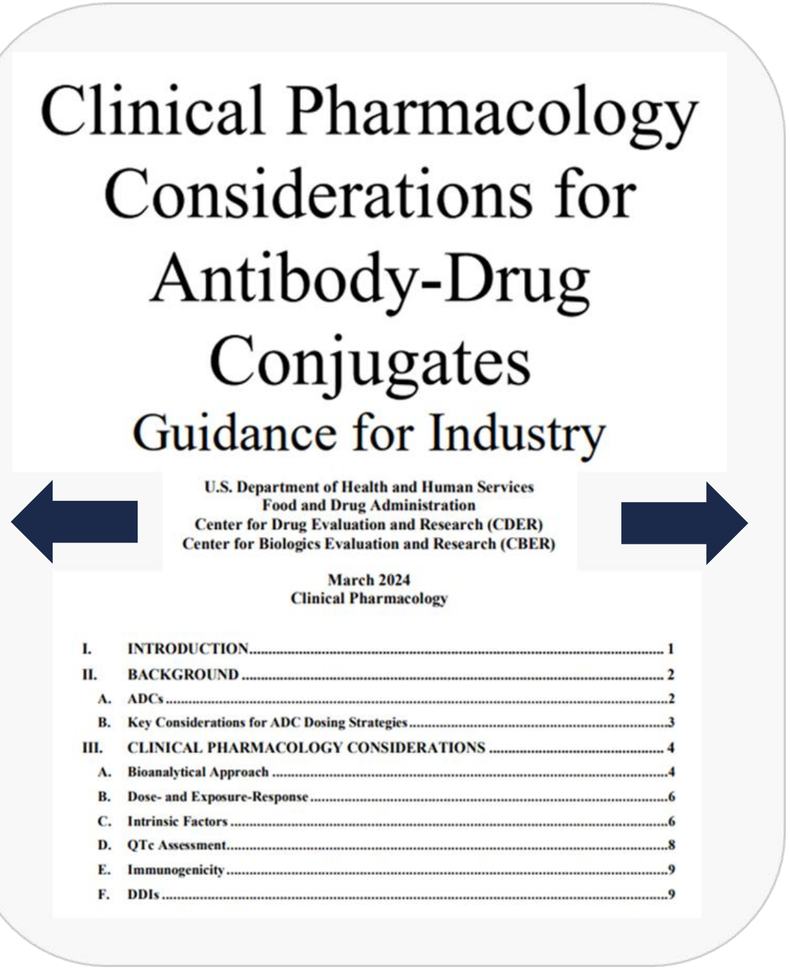
Exposure-response modeling to optimize dose before pivotal trials

2 Safety Assessment

QTc modeling, organ impairment, and DLT characterization

3 Population Factors

Intrinsic/extrinsic factors (Like DDI, Pgx)



Clinical Pharmacology Considerations for Antibody-Drug Conjugates

Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

March 2024
Clinical Pharmacology

I. INTRODUCTION 1
II. BACKGROUND 2
A. ADCs 2
B. Key Considerations for ADC Dosing Strategies 3
III. CLINICAL PHARMACOLOGY CONSIDERATIONS 4
A. Bioanalytical Approach 4
B. Dose- and Exposure-Response 6
C. Intrinsic Factors 6
D. QTc Assessment 8
E. Immunogenicity 9
F. DDIs 9

WHEN & WHAT DURING DRUG DEVELOPMENT

Preclinical	Nonclinical data & PK modeling
Ph 1/2	DE/optimisation/DLT Dose-response & biomarkers
Ph 3	Confirmatory ER analysis
Post-Market	Real-world optimization

Lilly's ADC PKPD strategy is fully aligned with the FDA's vision for model-informed dose optimization

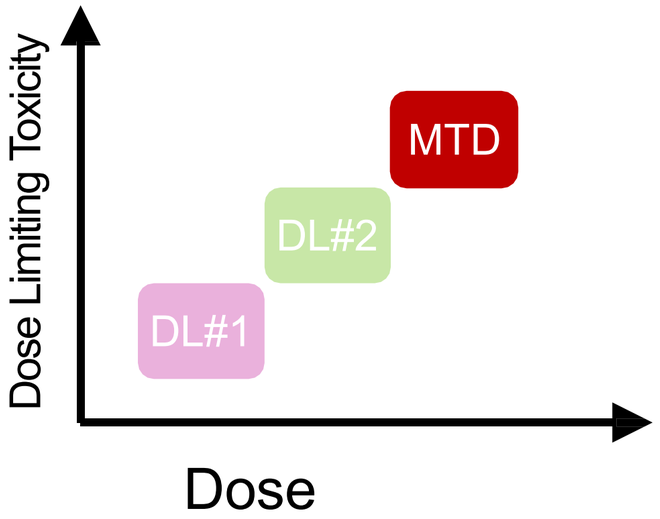
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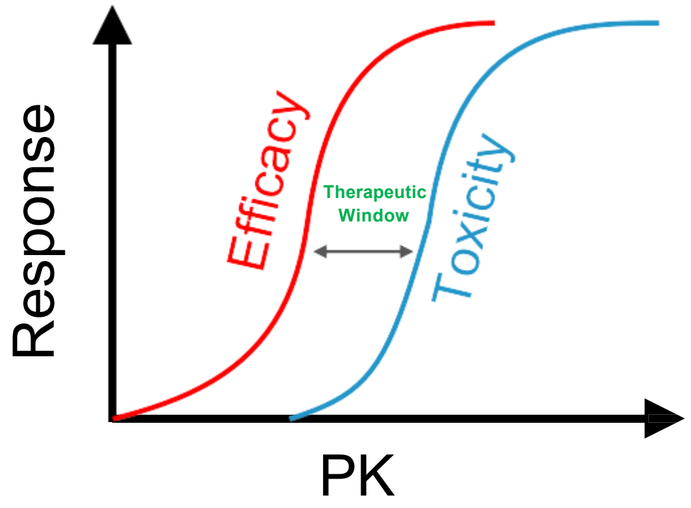
Clinical Utility Index (CUI) Modeling: Integrates PK, Efficacy, and Safety into One Comprehensive Measure for Optimum Biological Dose (OBD)

One Index. Three Dimensions. Smarter Dose Selection Decisions

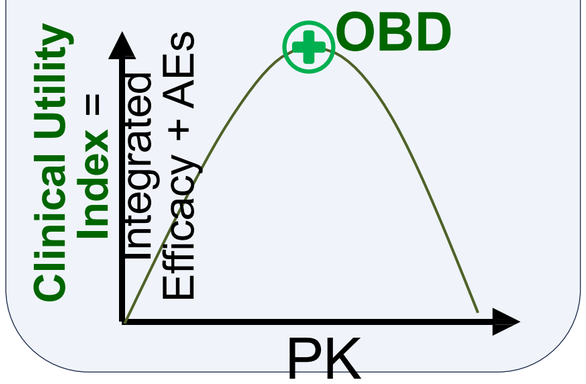
MTD approach ✕



Balance efficacy and safety ✕



Clinical Utility Index



Integrates PK exposure with multiple clinical outcomes

Quantifies benefit-risk across dose levels

Clinical Utility Index Transforms dose selection from a **single-dimension toxicity/efficacy question** into a **comprehensive benefit-risk optimisation**

CUI: Clinical Utility Index; MTD: Maximum Tolerable Dose ; OBD: Optimum Biological Dose; PK =Pharmacokinetics

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CUI Modeling: From Endpoints to Optimal Biological Dose

1 COLLECT ENDPOINTS
PK, ORR, AEs across DLs

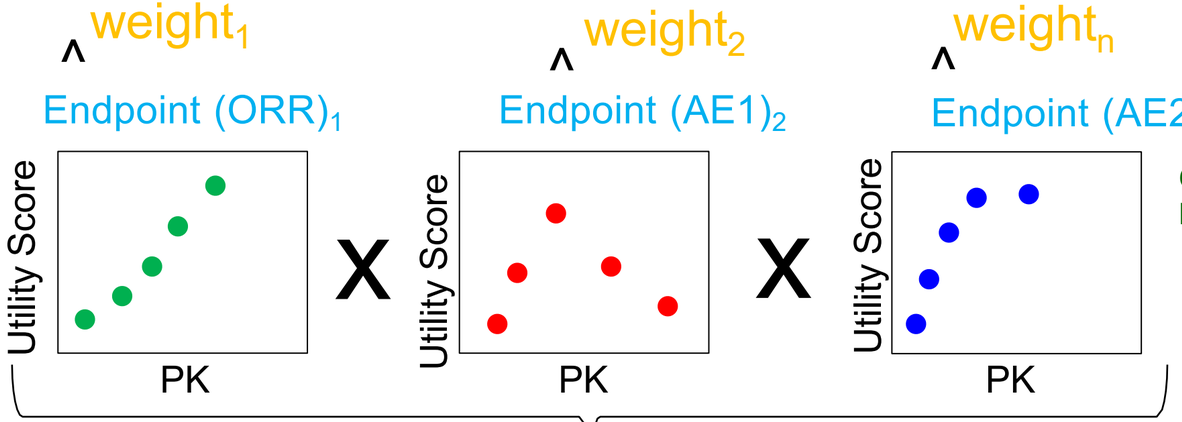
2 ASSIGN WEIGHTS
Assign weights to each endpoint based on therapeutic context

3 COMPUTE CUI SCORE
CUI = Σ (weight$_i$ × utility$_i$)
Single composite score per dose level

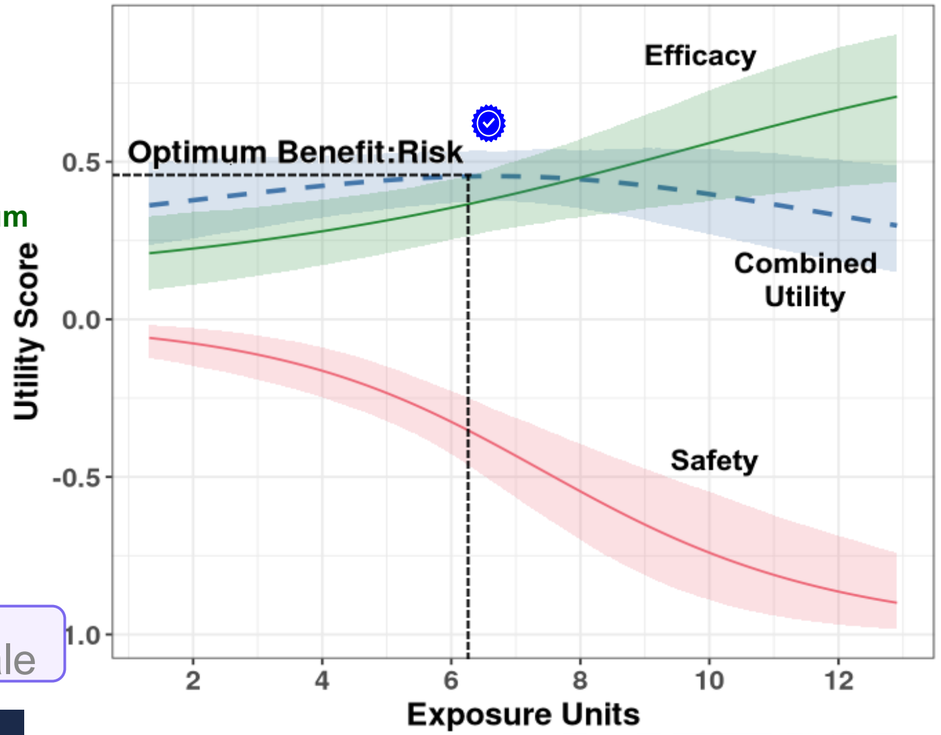
4 IDENTIFY OBD
Dose with maximum CUI score = OBD for pivotal trials

OUTPUT: Single CUI Score Per Dose

Dose Level 1 (Low)	CUI = 0.42
Dose Level 2 (Mid)	**CUI = 0.78 ★ OBD**
Dose Level 3 (High)	CUI = 0.65
Dose Level 4 (MTD)	CUI = 0.31



CUI for OBD & Optimum Benefit-Risk



$$\mathrm{CUI}(x) = \prod_{i=1}^{n} E(x)_i^{w_i} \times \prod_{j=1}^{m} \left(1 - AE(x)_j\right)^{w_j}$$

CUI = Σ (weight$_i$ × utility$_i$) where each endpoint is mapped to a 0–1 utility scale

Weighted combination of PK, efficacy, and safety endpoints into a single utility score to identify the Optimum Biological Dose

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CUI: Clinical Utility Index; OBD: Optimum Biological Dose; PK =Pharmacokinetics; ORR: Overall Response Rate; AE=Adverse Event

Growing Visibility & Democratisation of CUI Methodology Across the Globe

WHY IT MATTERS: Democratizing CUI at Peer-reviewed journals/Conferences establishes CUI as a credible, reproducible framework recognised by the Global **regulators**, Clinicians and industry

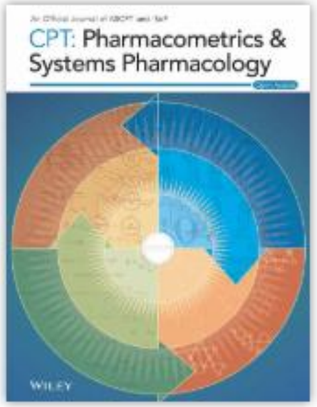
PUBLICATIONS

Postmarketing Assessment of Antibody–Drug Conjugates: Proof-of-Concept Using Model-Based Meta-Analysis and a Clinical Utility Index Approach

Innocent Gerald Asiimwe, Nour Chtiba, Samer Mouksassi, Goonaseelan (Colin) Pillai, Raimund M. Peter, Eunice Yuen, Venkatesh Pilla Reddy

First published: 04 March 2025 | https://doi.org/10.1002/psp4.70013

CPT: Pharmacometrics & Syst Pharmacol (2025)



FDA-AACR WORKSHOP

FDA AACR

FDA-AACR Public Workshop On

OPTIMIZING DOSAGES FOR ONCOLOGY DRUG PRODUCTS: QUANTITATIVE APPROACHES TO SELECT DOSAGES FOR CLINICAL TRIALS

February 15 and 16, 2024 | Washington, DC

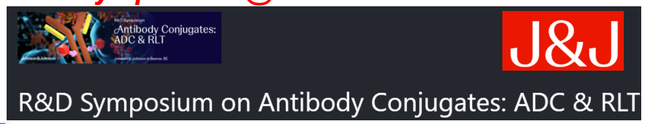
Clinical Utility Index (CUI): a Tool to Support Dose Selection as an Alternative to Maximum Tolerated Dose (MTD)

CONFERENCES & INDUSTRY

CUI is **gaining traction** as the **industry standard for ADC dose optimisation** — endorsed by FDA and pharma leaders

ESMO

R&D Symposium @ Johnson &Johson



15th Annual WORLD adc LONDON 2025

Exposure-Response–Based Multiattribute Clinical Utility Score Framework to Facilitate Optimal Dose Selection for Oncology Drugs

Authors: Yiming Cheng, PhD, Shuyu Chu, PhD, Jie Pu, PhD, Min Chen, PhD, Kevin Hong, PhD, Paulo Maciag, MD, PhD, Ivan Chan, PhD, Li Zhu, PhD, Akintunde Bello, PhD, and Yan Li, PhD

AUTHORS INFO & AFFILIATIONS

J Clin Oncol 42, 4145-4152(2024) • Volume 42, Number 35 • DOI: 10.1200/JCO.24.00349

PAGE

pharmacometrics africa

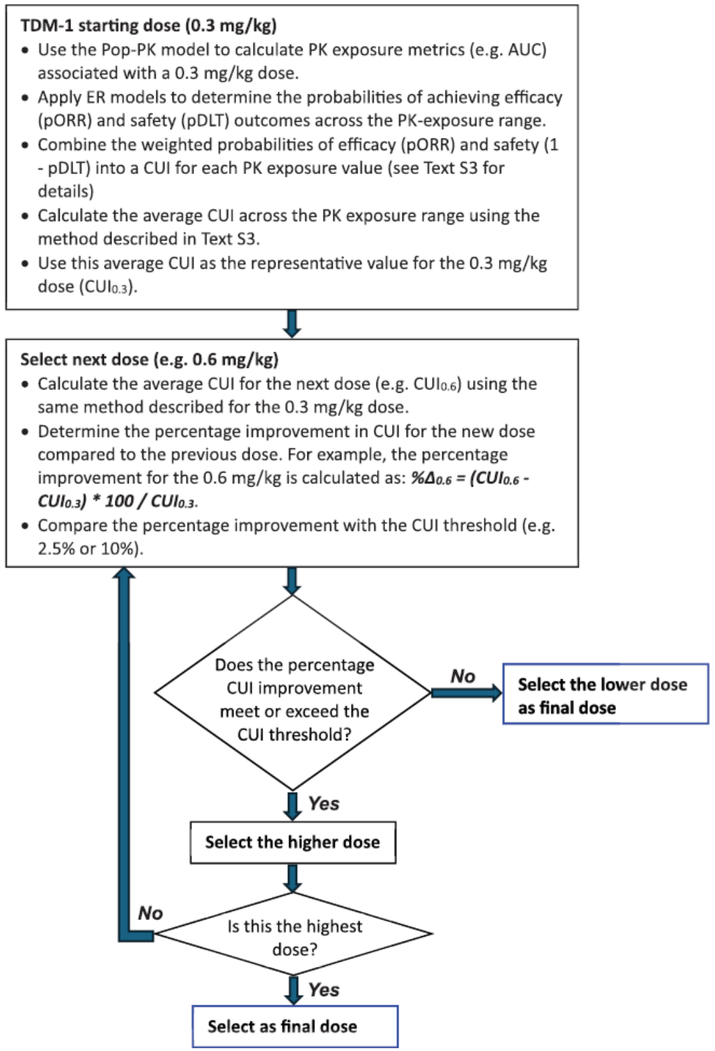
ISoP INTERNATIONAL SOCIETY OF PHARMACOMETRICS

Pharmacokinetics (U.K.) PK

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CUI Workflow using Enhertu and Kadcyla ADCs



Pop-PK → Simulations → Exposure-Response Analysis → CUI

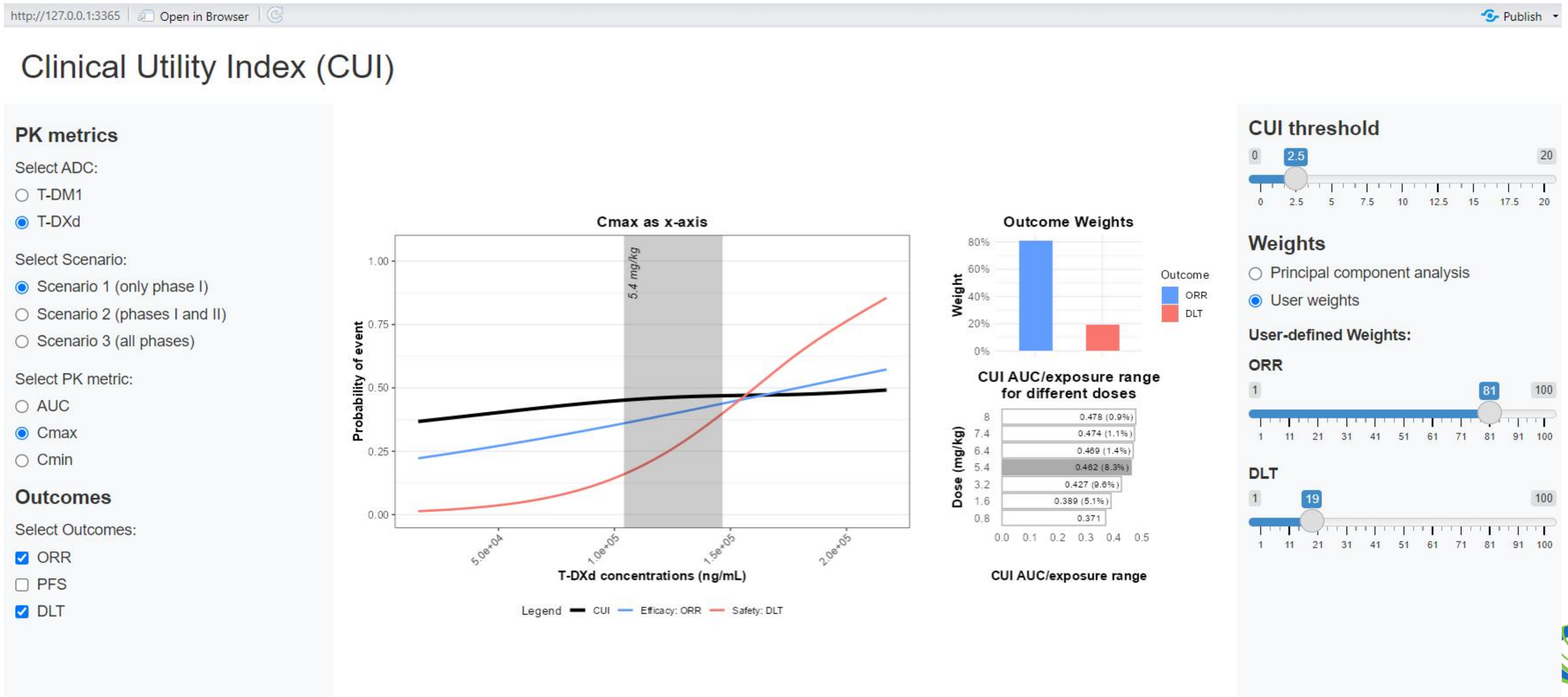
GitHub

github.com/iasiimwe/adc_cui

- .gitignore
- 1. PopPK analysis.R
- 2. Simulating with uncertainty.R
- 3. ER analysis.R
- 4. CUI analysis.R
- 5a. Process Shiny data.R
- 5b. CUI_shiny.R
- 6. TDXd.R
- README.md
- adc_cui.Rproj
- base.txt

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CUI Interactive Shiny app is designed specifically to empower clinical and medical teams (pilot version)



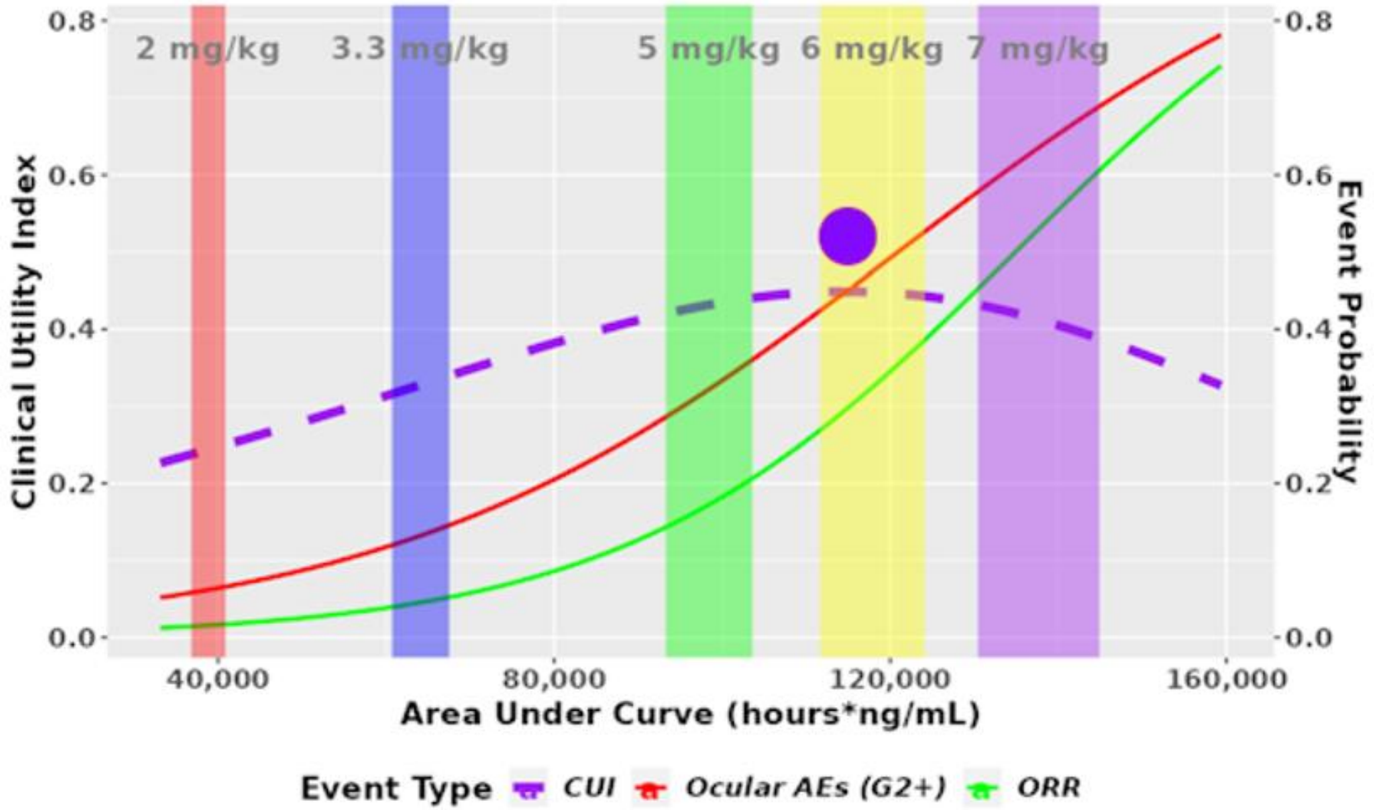
CUI can be easily applied across drug development stages

2026

CUI Validation Using 8 Approved ADCs and >100 Clinical Trial Data

- Enhertu
- Elahere (FRa)
- Zynlonta
- Kadcyla
- PADCEV
- Tivdak
- Trodelvy
- Torv-S (FRa)

CUI for Mirvetuximab Soravtansine (Elahere; FRa-ADC) using publicly available **mean** data



FRa: Folate Receptor α; CUI: Clinical Utility Index; OBD: Optimum Biological Dose; PK =Pharmacokinetics

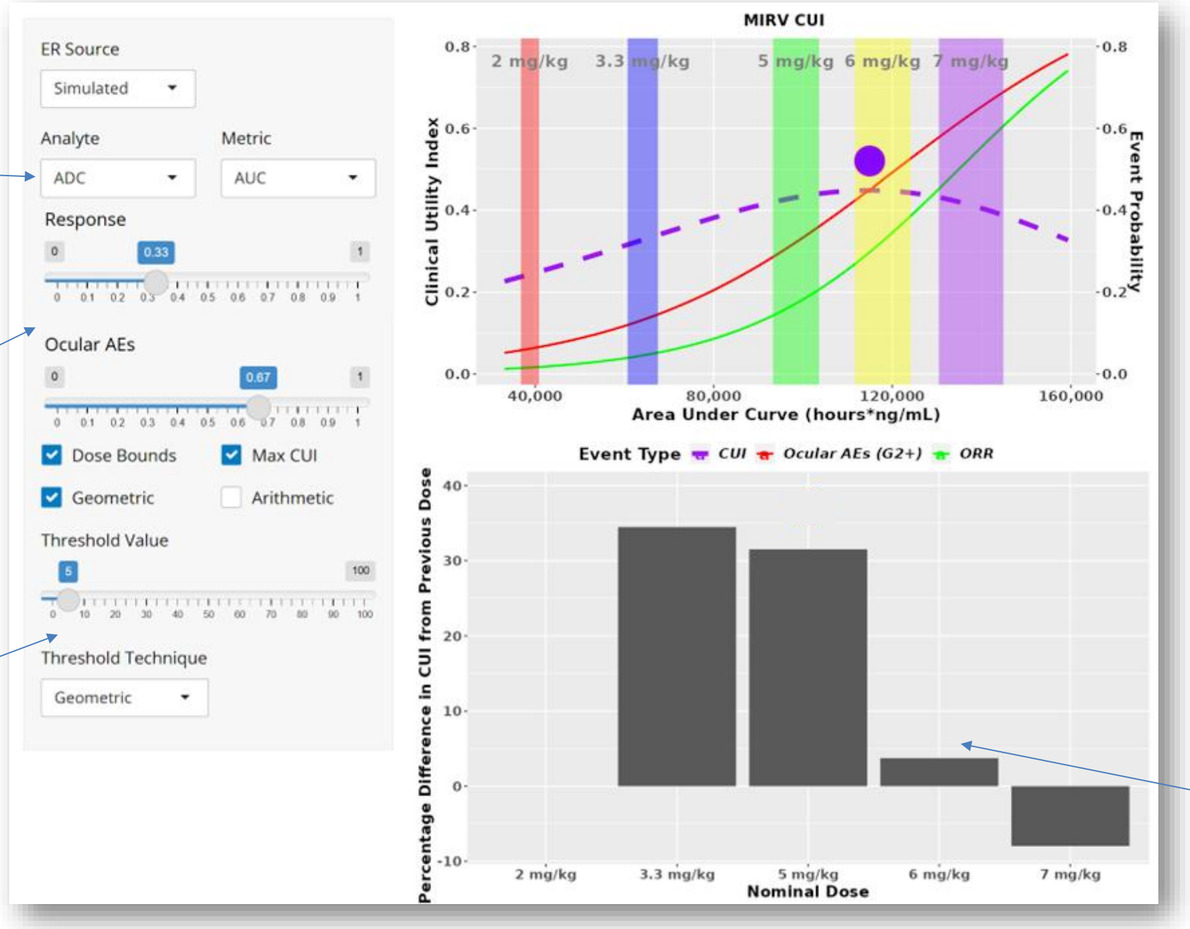
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Pilot CUI to Implementation Stage at Lilly using Approved ADC Mirvetuximab Soravtansine (mean data)

Variable Analytes and Exposure Metrics

Weighting Selection

Experimental Thresholding



Aggregated Geometric CUI

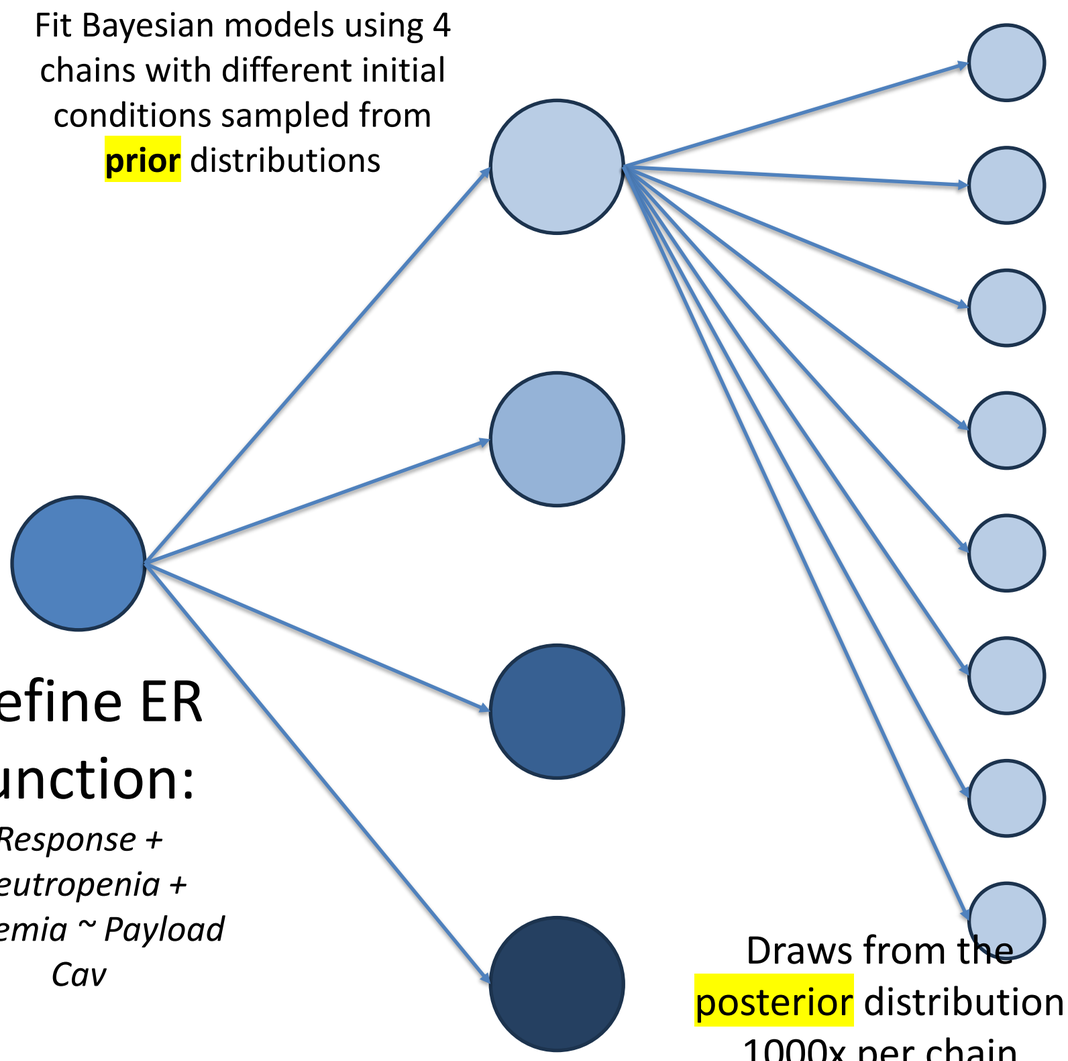
Dose	Event	Count	N	Count/N
2	Neutropenia	1	30	0.033333
3	Neutropenia	1	11	0.090909
4	Neutropenia	5	29	0.172414
6	Neutropenia	9	16	0.5625
2	Anaemia	4	30	0.133333
3	Anaemia	0	11	0
4	Anaemia	5	29	0.172414
6	Anaemia	10	16	0.625
2	Response	2	10	0.2
3	Response	3	8	0.375
4	Response	6	12	0.5
6	Response	3	4	0.75

Simulated Conc. Bounds

Optimal Dose

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Moving from Mean to individual level data: The individualised Bayesian CUI (iBCUI) to account for uncertainty and variability



Fit Bayesian models using 4 chains with different initial conditions sampled from **prior** distributions

Define ER function:

Response + Neutropenia + Anaemia ~ Payload Cav

Draws from the posterior distribution 1000x per chain

Predict an exposure-response curve for each dependent variable, for each draw (n=4000)

Calculate a CUI for each draw, then summarise across all CUI predictions

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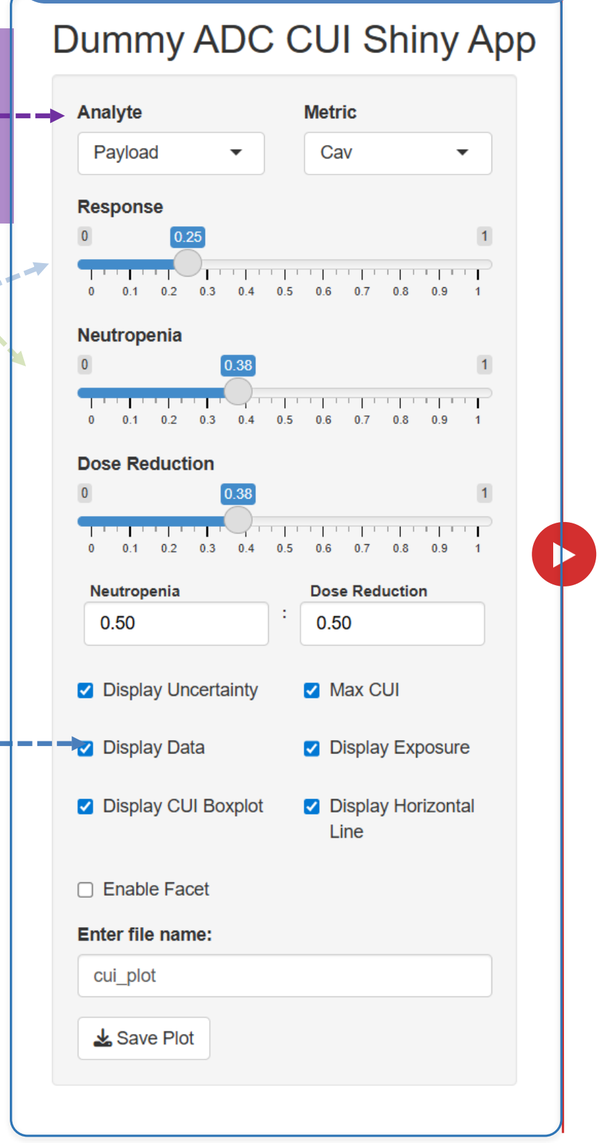
CUI Interactive App (V.4): For Optimum Ph3 Dose Selection

ADC Analytes & PK Exposure Metrics

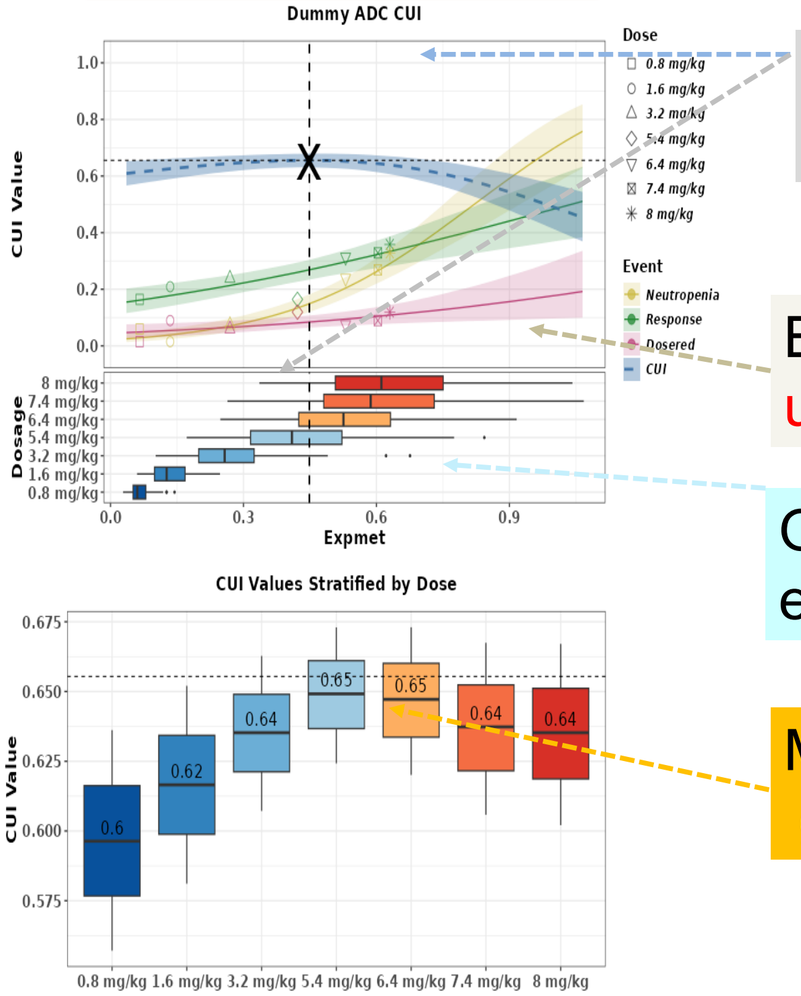
Weight Selection

Plotting Options

INPUT: Dummy ADC



OUTPUT: CUI Analysis



Individual binary data with or without events

Bayesian approach to account for uncertainty/patient variability

Observed (or simulated) PK exposure bounds

Median CUI per nominal dose group

FRa: Folate Receptor α; CUI: Clinical Utility Index; X: Optimum Biological Dose; Cavg =Average Concentration using AUC/dosing interval; PK =Pharmacokinetics

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One framework, two strategic applications across the ADC lifecycle

PRECLINICAL	PHASE 1	PHASE 2	PHASE 3	MARKET

(Pre)-Clinical Utility Index ◀ *CUI informs decisions across the entire development continuum* ▶

WITHIN AN ADC — Optimize the Molecule

1. Identify optimal RP2/3 Dose
2. Utility-informed trial design & decision-making at the team level
3. Benefit-risk characterization (TI)
4. Comparison of different formulations
5. Cancer type, gender, age, PGx

BETWEEN ADCs — Differentiate vs. Competition

1. Compare ADC candidates using a utility score to select the best molecule
2. Understand positioning & differentiation from market leader
3. Identify which endpoints and attributes drive clinical and commercial value

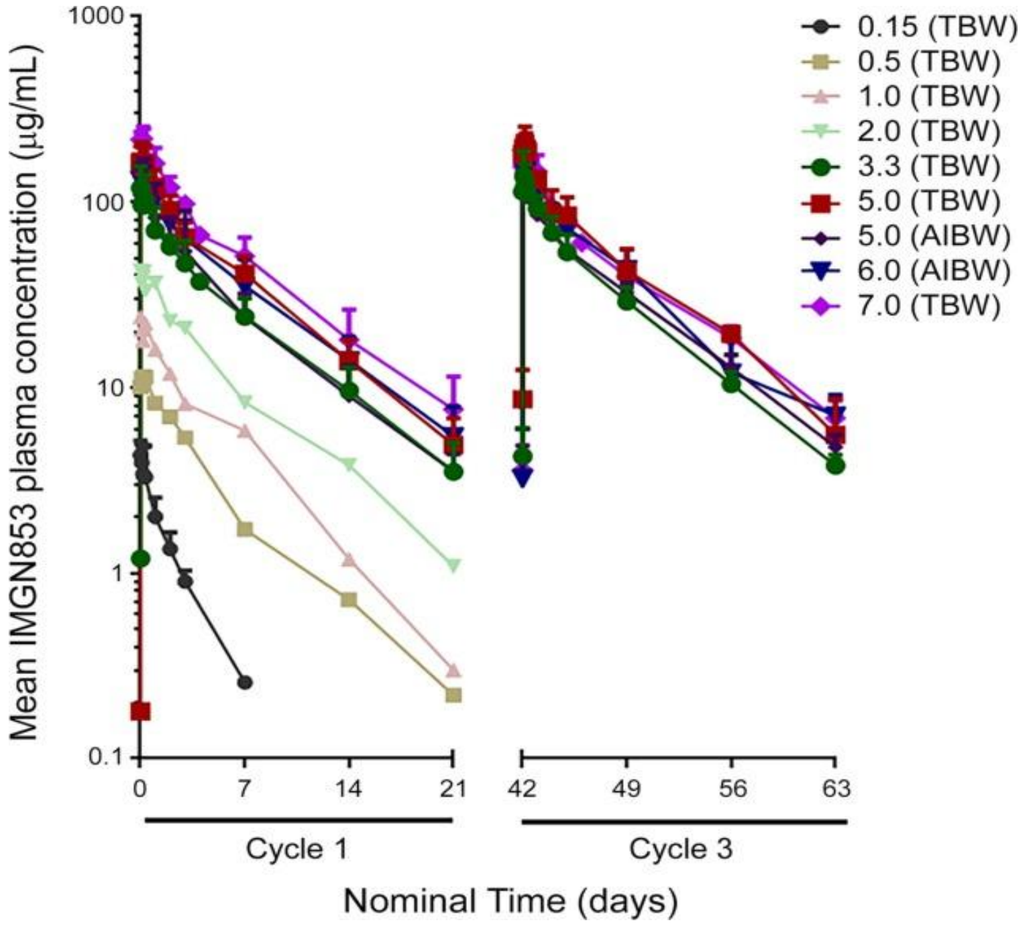
KEY TAKEAWAY: One score unifies dose optimization within a molecule and competitive differentiation between ADCs

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CUI Case Study Mirvetuximab Soravtansine PK and Clinical Data

CUI Steps: PK Modeling → Simulations → Exposure- Response Analysis → CUI



Indication: *ovarian, fallopian tube, or primary peritoneal cancer*

MTD/RP2D: *6 mg/kg AIBW Q3W*

- Following IV infusion, the exposures (Cmax and AUC0-INF) of ADC and total Ab showed dose-proportional increase at 3.3 to 7 mg/kg total BW
- Recommended dosing regimen: IV 6 mg/kg Q3W adjusted ideal body weight (AIBW)

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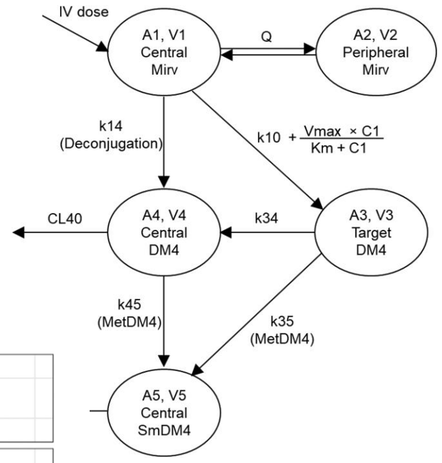
CUI: Clinical Utility Index; PK =Pharmacokinetics

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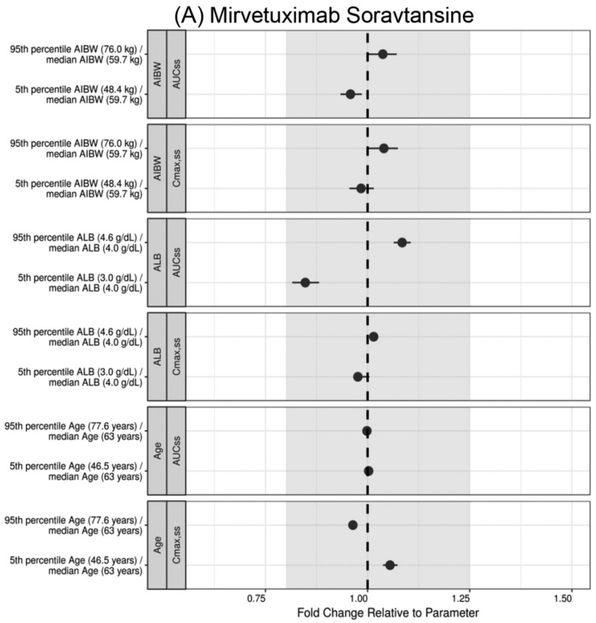
PK Model accounts for all 3 analytes to facilitate CUI building

CUI Steps: PK Modeling → Simulations → Exposure- Response Analysis → CUI

PK model



Covariate model



KEY FINDINGS

WHAT PK MODEL DESCRIBES

3-analyte PK model for conjugated ADC, DM4 and metabolite of DM4

COVARIATE EFFECTS/PATIENT VARIBILITY

Forest plot quantifies the **impact of patient factors** (body weight, albumin, LFT) on PK parameters for robust ER analysis. No impact except the body weight and Albumin

CLINICAL IMPLICATION

PK model feeds directly into the CUI framework — enabling **simulation of exposure** profiles across doses for CUI

BOTTOM LINE: PK modeling provids the **exposure foundation** for CUI-based optimal dose selection of Mirv-S

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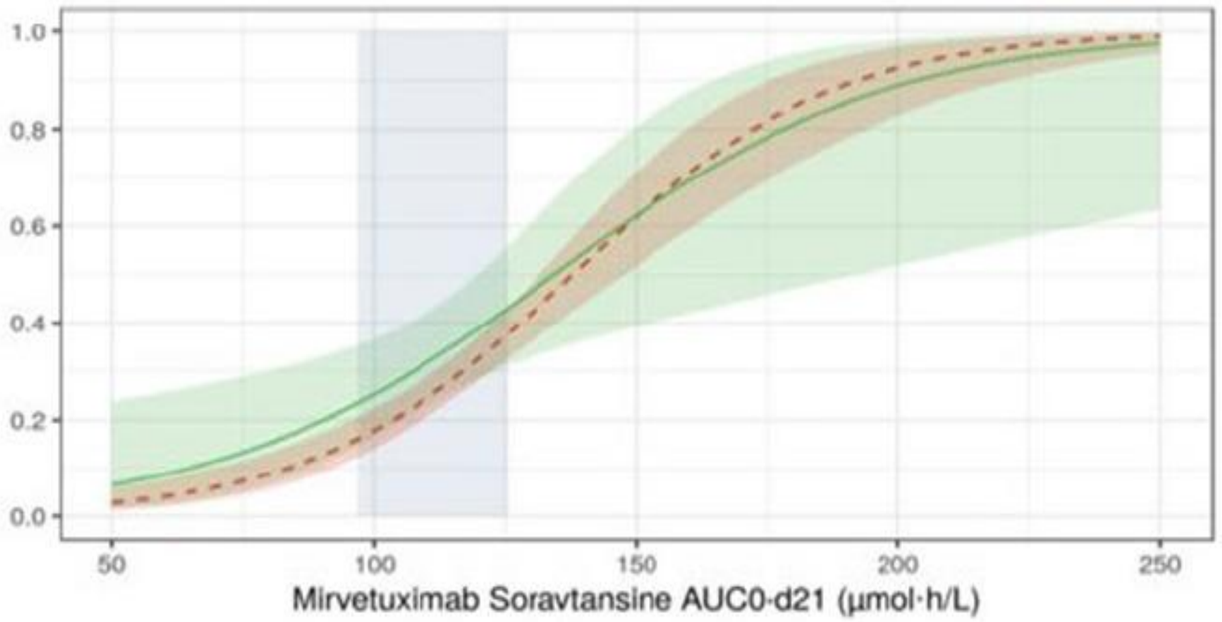
PK =Pharmacokinetics, ER: Exposure Response Analysis; DAR= Drug Antibody Ratio; FRa: Folate Receptor α

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Conventional Exposure-Response to select the Ph3 Dose for Mirv-S?

CUI Steps: PK Modeling → Simulations → Exposure-Response Analysis → CUI

Green: ORR
Red: Ocular AEs
Grey: Predicted 6 mg/kg AIBW Q3W exposure



Safety concern*: ocular toxicity in 59% patients, including 8% Grade 3*

E-R relationship for efficacy*: higher AUC, higher ORR*

E-R relationship for safety*: higher C_{max}/AUC, higher ocular toxicity*

Key Insight: CUI is essential for determining the optimal dose, as different AEs correspond to distinct PK thresholds

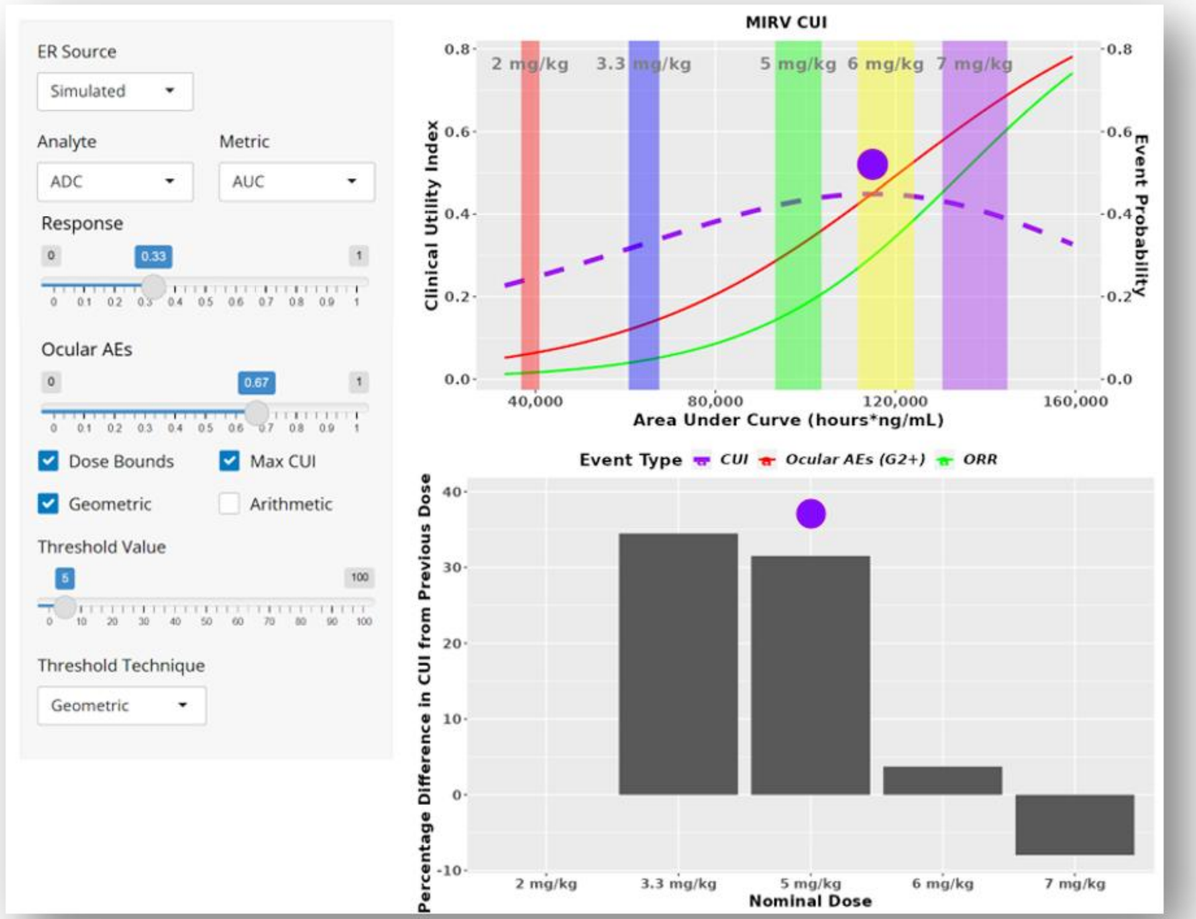
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PK =Pharmacokinetics; ORR: Overall Response Rate

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CUI Analysis (blinded) using confirmed ORR Indicated 6 mg/kg dose as optimal dose for Mirv-S

CUI Steps: PK Modeling → Simulations → Exposure-Response Analysis → CUI



CUI Analysis confirmed 6 mg/kg dose

CUI enables rapid integration of new data to evaluate its impact on dose choices

DE: Dose Escalation; DO: Dose Optimization; CUI: Clinical Utility Index; OBD: Optimum Biological Dose; PK =Pharmacokinetics; cORR: confirmed Overall Response Rate

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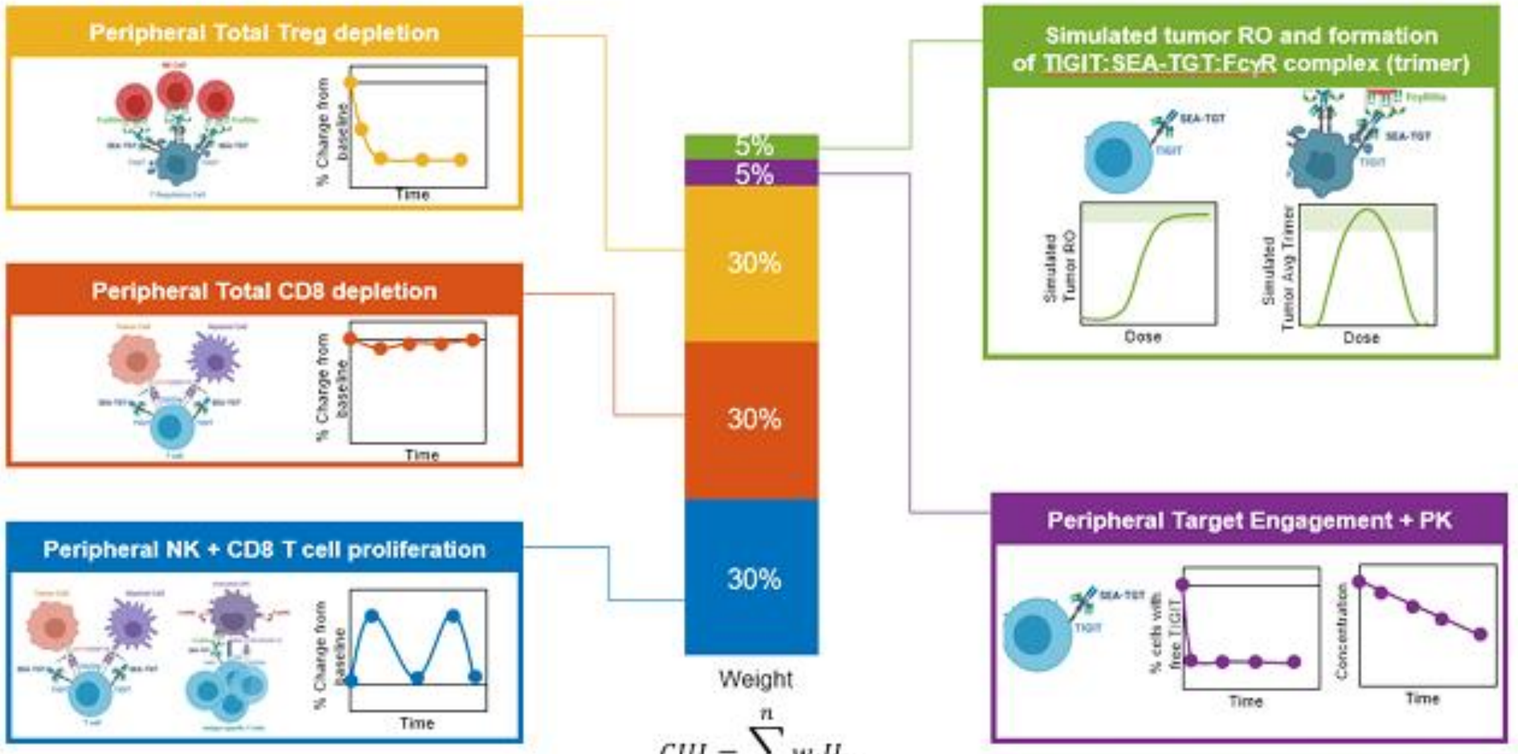
Demonstration of Clinical Utility Index Shiny App using Mirv-S ADC to confirm Ph3 dose

Link: ADC CUI Shiny

Early Clinical Development Stage CUI Example

CUI to Compare Biological Activity Across Dose Cohorts

All selections were prespecified using SEA-TGT preclinical and literature-based data to limit bias. Endpoint weights were based on *a priori* consensus that balanced relevant biological activity with variability and/or uncertainty in output.



$$CUI = \sum_{i=1}^{n} w_i U_i$$

CUI is the sum of the weighted (w) average utility functions (U) for all endpoints of interest (i)

Patilea-Vrana et al. Cancer Res 2023;83(7_Suppl):Abstract nr 5668.

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Acknowledgements



Daniel Bending, Post-Doc, Eli Lilly, UK)



Innocent Asiimwe, Liverpool University (APT Fellow), now at Novartis, UK



Eunice Yuen, Global PK/PD & Pharmacometrics, Eli Lilly, UK

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CUI for quantitative benefit – risk assessment in Drug Development Decision – Non-Oncology

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CUI and other forms of multi-attribute utility approaches

Samer MOUKSASSI, PhD

Senior Director – Application Scientist
Accelerated Clinical Evidence Certara Radnor PA

ACoP 2026 Workshop; 15th October 2026

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Beyond oncology: where CUI has been applied

1. **Formulary choice — ranking five antipsychotics for schizophrenia.** A multi-attribute model weighted efficacy, adverse effects, cost, and adherence to pick a preferred agent.

2. **Dosing-algorithm choice — a new basal insulin in type 2 diabetes.** A clinical utility index combined four endpoints to choose between two dose-titration algorithms.

3. **Longitudinal example — a reformulation decision.** A longitudinal clinical utility index

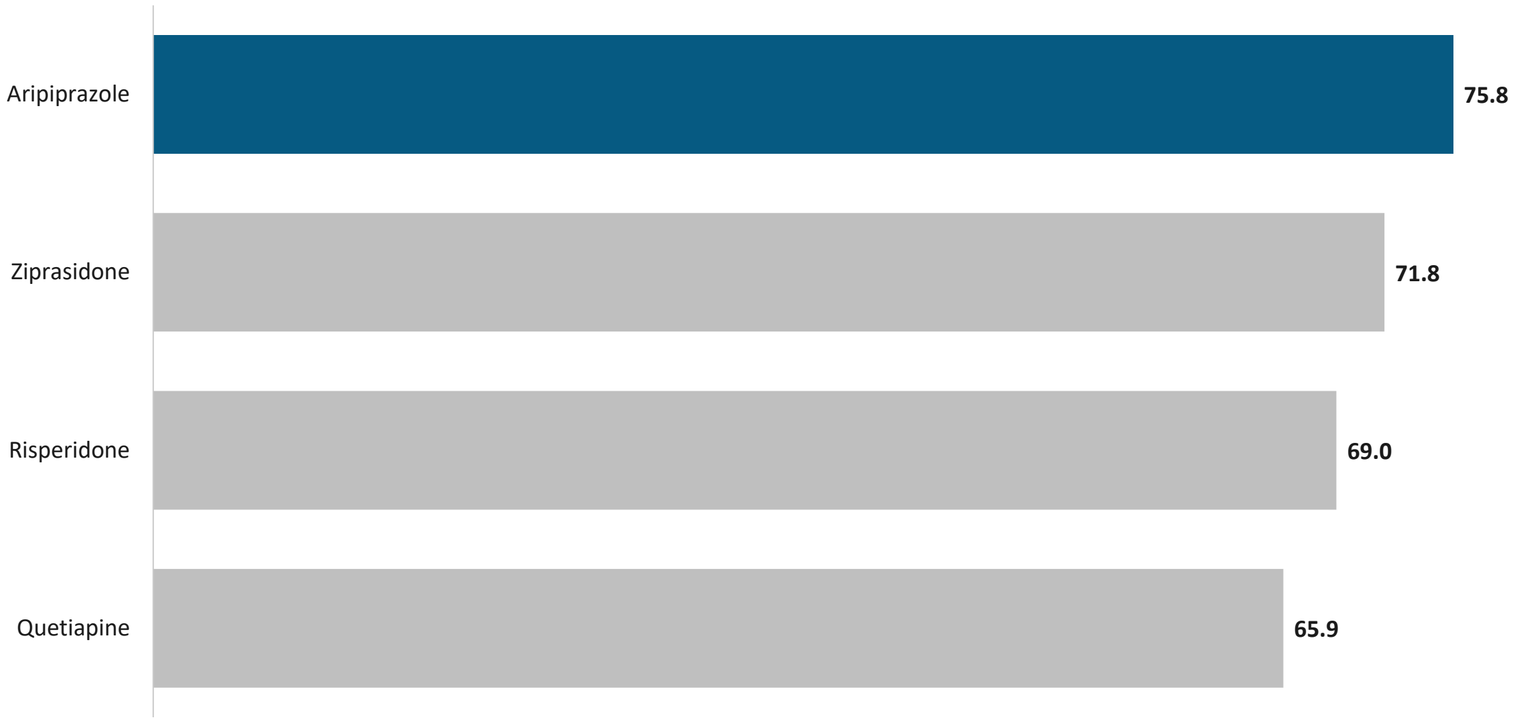
4. **Regulatory adoption — FDA and EMA now run the same math.** Agencies have started using multi-criteria decision analysis themselves to resolve benefit–risk disagreements.

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Formulary choice: a multi-attribute ranking of antipsychotics

Total utility score, 0–100 scale · MAUT model of 5 atypical antipsychotics



Attribute weights

Efficacy **35%**

Adverse effects **35%**

Cost **20%**

Adherence **10%**

Olanzapine scored lowest of the five agents (exact value not reported). A sensitivity analysis did not displace aripiprazole as the top-ranked agent.

Source: Bettinger TL, Shuler G, Jones DR, Wilson JP. Schizophrenia: multi-attribute utility theory approach to selection of atypical antipsychotics. Ann Pharmacother. 2007;41(2):201–207.

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Formulary choice: a multi-attribute ranking of antipsychotics

Table 3. Adverse Effects: Factor Values[a] and Utility Scores

Adverse Effect	Risperidone	Olanzapine	Quetiapine	Ziprasidone	Aripiprazole
NMS	25	25	25	25	25
EPS	50	25	0	25	25
Prolactin	50	25	0	25	0
Weight gain	50	75	50	25	25
Glucose	25	50	25	0	0
Lipids	25	50	25	0	0
QTc interval prolongation	25	0	0	50	0
Sedation	25	25	50	0	25
Hypotension	25	50	50	25	25
Anticholinergic	0	25	0	0	0
Utility score[b]	60.0	53.3	70.0	76.7	83.3

EPS = extrapyramidal symptoms; NMS = neuroleptic malignant syndrome.
[a]Values were assigned as follows: 0 = negligible or no risk, 25 = low risk, 50 = moderate risk, 75 = high risk, and 100 = very high risk of the adverse effect at a therapeutic dose.
[b]All factors are given equal weight to calculate the utility score.

Table 4. Estimated Yearly Medication Acquisition and Laboratory Costs[23]

Cost	Risperidone	Olanzapine	Quetiapine	Ziprasidone	Aripiprazole
AWP ($)	3379.90	5675.75	5639.25	3920.10	4033.25
Laboratory ($)	96.17	95.21	64.35	102.59	64.35
Utility score	86.8	62.3	64.0	81.2	80.5

AWP = average wholesale price.

score (U_T). The following equation was used to calculate the total utility score for each atypical antipsychotic:

$$U_T = [W \times UE] + [W \times UAE] + [W \times UC] + [W \times UA] \quad \text{Eq. 3}$$

where U_T = total utility score, W = attribute weight, U_E = efficacy utility score, U_{AE} = adverse effects utility score ($U_{AE} = \Sigma [U_f \times W]$ adverse effects), U_C = cost utility score ($U_C = \Sigma [U_f \times W]$ cost), and U_A = adherence utility score.

The alternative with the highest total utility score was considered the preferred agent.

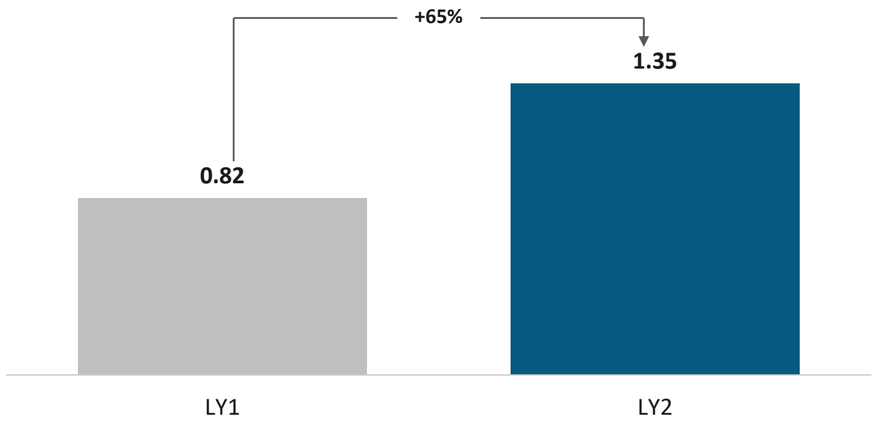
SENSITIVITY ANALYSIS

The last step in the MAUT analysis included conducting a sensitivity analysis. Because weighting each of the attributes was a subjective process, a sensitivity analysis was

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Dosing-algorithm choice: a new long-acting insulin

Multiplicative clinical utility index · LY2605541 dosing algorithms · phase 2, type 2 diabetes



CUI components

- HbA1c
- % of patients reaching HbA1c ≤7%
- Hypoglycemia rate
- Time to steady-state dose

3,000-sample bootstrap: 48% probability LY2 outperforms LY1 on the combined index, versus 28% the reverse.

Source: Manner DH, Luo J, Qu Y, Berry S, Gaydos B, Jacober SJ. A clinical utility index for selecting an optimal insulin dosing algorithm for LY2605541 in patients with type 2 diabetes pretreated with basal insulin. Diabetes Technol Ther. 2014;16(8):499–505.

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Dosing-algorithm choice: a new long-acting insulin

Multiplicative clinical utility index · LY2605541 dosing algorithms · phase 2, type 2 diabetes

TABLE 2. CLINICAL UTILITY INDEX COMPONENTS AND COMPUTED VALUES FOR TWO LY2605541 DOSING ALGORITHMS RELATIVE TO INSULIN GLARGINE DOSING

Component	*LY1 (n = 98)*[a]	*LY2 (n = 97)*[a]	*GL (n = 93)*[a]	*CUI relative to GL*	
				For LY1	*For LY2*
At 12 weeks					
HbA1c (%) LS mean change from baseline (SE)	−0.64 (0.07)	−0.83 (0.07)	−0.64 (0.08)	1.00	1.23
Number of patients reaching HbA1c < 7% (%)	47 (55.3%)	47 (51.7%)	44 (48.4%)	1.17	1.08
Hypoglycemia rate (events/patient/30 days) during 12 weeks: LS mean (SE)	1.74 (0.33)	1.59 (0.31)	1.68 (0.36)	0.91	1.13
Time when 50% of patients reached steady-state dose (weeks)	5.00	4.00	3.14	0.77	0.89
CUI (Components 1–4)	LY1/GL, 0.82; LY2/GL, 1.35 CUI probabilities for LY2 > LY1, LY2 = LY1, and LY2 < LY1 are 48%, 24%, and 28%, respectively.[b]				
CUI (Components 1–3)	LY1/GL, 1.07; LY2/GL, 1.51 CUI probabilities for LY2 > LY1, LY2 = LY1, and LY2 < LY1 are 46%, 24%, and 30%, respectively.[b]				

This table was previously presented in an abstract by Manner et al.[14]
[a]Bergenstal et al.[12]
[b]Based on 3,000 bootstrap samples.
CUI, clinical utility index; GL, insulin glargine; HbA1c, glycosylated hemoglobin; LS, least-squares; LY1, dosing algorithm 1 for LY2605541; LY2, dosing algorithm 2 for LY2605541.

To compute the CUI for each LY2605541 algorithm, the four individual component utilities were multiplied, providing a quantitative assessment of the relative benefits and risks of the dosing algorithms. Because of the multiplicative function, if any one component had a utility value of 0 (indicating no value), then the CUI had a value of 0, which was a desirable property. If one of the components had a utility value of <1 (indicating a less than desirable outcome), then the CUI value was lowered. For example, if the utility value for hypoglycemia rate was 0.5, then the CUI value was decreased by 50%.

Source: Manner DH, Luo J, Qu Y, Berry S, Gaydos B, Jacober SJ. A clinical utility index for selecting an optimal insulin dosing algorithm for LY2605541 in patients with type 2 diabetes pretreated with basal insulin. Diabetes Technol Ther. 2014;16(8):499–505.

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The Dopahexadine case: a reformulation decision, not a go/no-go

1. **The setup —** a mature, marketed drug for a chronic neuromuscular condition; the question was whether reformulating it could improve the benefit–risk balance, not whether to launch it.

2. **The team —** cross-functional: Clinical Development, Project Management, Commercial, Biostatistics and PK, led by a consulting pharmacometrician trained in CUI.

3. **The process —** four steps — determine the attributes that matter, define response levels within each, quantify relative preference within an attribute, then assign weights across attributes.

4. **The check —** weights were elicited twice — once from the **project team**, once from a **physician survey** — to see where clinical judgment and prescriber preference agreed and where they diverged.

Elicitation of the attributes and weights

3

The process — four steps

determine the attributes
define response levels within each
quantify relative preference across those levels
assign weights across attributes.

When weights differ run under various assumptions— the analysis was repeated under each set as a sensitivity check; the reformulation call held up either way.

Two attributes were tied out, when data is insufficient — Elevated LFTs and Compliance lacked the data to tell formulations apart

Weights started with the team, then a physician survey checked them — rankings shifted — Elevated LFTs moved from 2nd (team) to 6th (physicians) — but the attribute weights themselves stayed close.

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Every endpoint was translated into a 0-1 utility score before it was weighted

Endpoint	Worst	Middle	Best
Efficacy (Δ SES)	≥ -0.3 (not meaningful)	-0.6 to -0.3 (u = 0.60)	≤ -0.6 (excellent)
Hypotension (Δ SBP)	-30 to -15 mmHg	-15 to -5 mmHg (u = 0.70)	-5 to 0 mmHg
Drowsiness (% incidence)	52% - 78%	26% - 52% (u = 0.60)	0% - 26%
Duration of effect (h)	0 - 2 h	2 - 5 h (u = 0.50)	5 - 12 h
Dyskinesia (% incidence)	35% - 40%	15% - 35% (u = 0.70)	0% - 15%

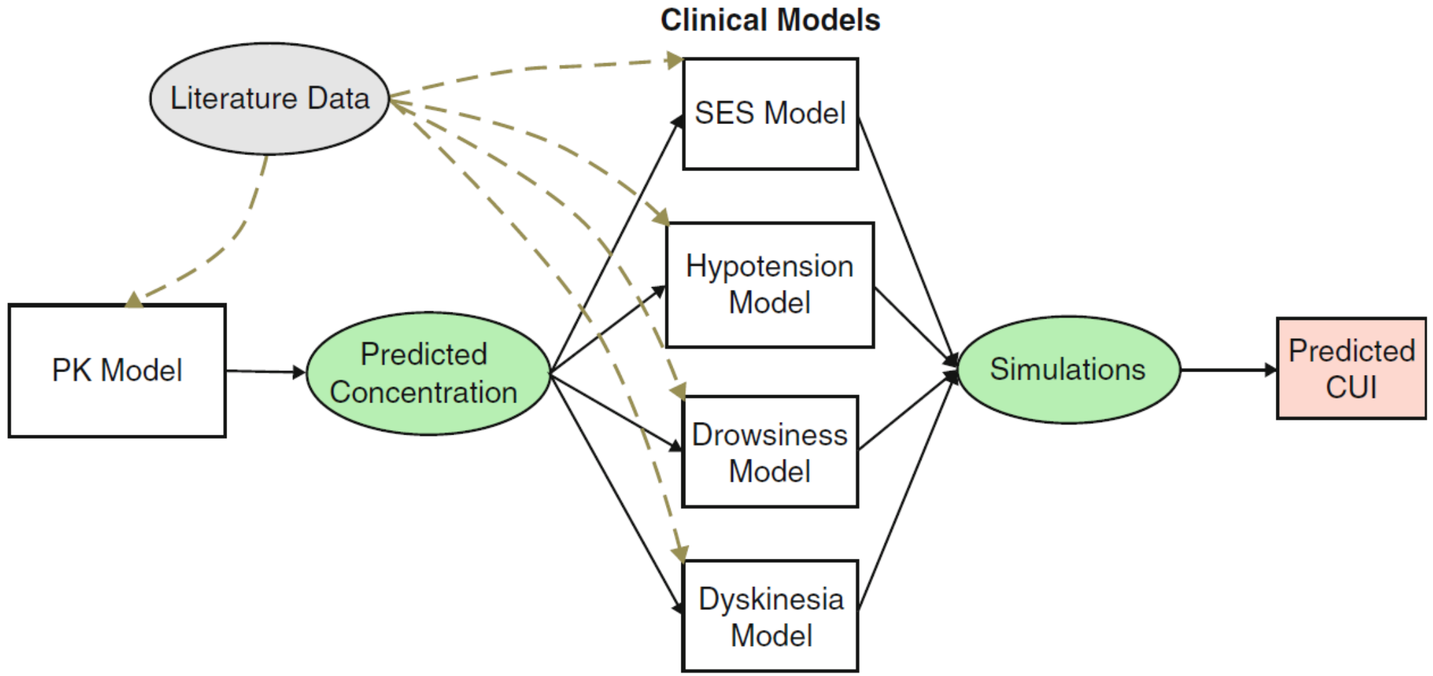
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Physicians and the team agreed on efficacy, but diverged sharply on liver-function risk

Attribute	Physician rank	Physician weight	Team rank	Team weight
Efficacy (Δ SES)	1	0.228	1	0.207
Compliance (%)	2	0.169	5	0.120
Hypotension (Δ SBP)	3	0.139	4	0.140
Drowsiness (% incidence)	4	0.135	3	0.154
Duration of effect (h)	5	0.112	7	0.098
Elevated LFTs (% >3x ULN)	6	0.109	2	0.182
Dyskinesia (% incidence)	6	0.109	6	0.101

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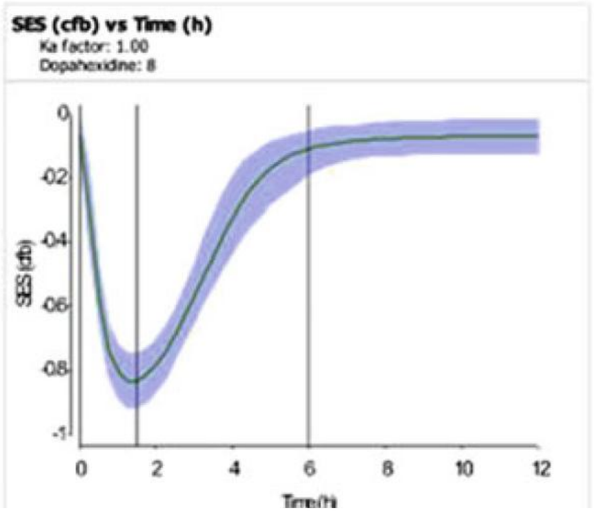
PK and exposure-response models fed simulations that predicted the CUI

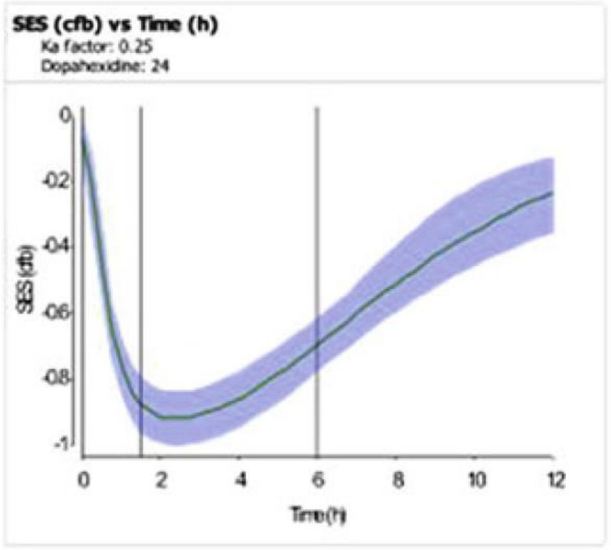


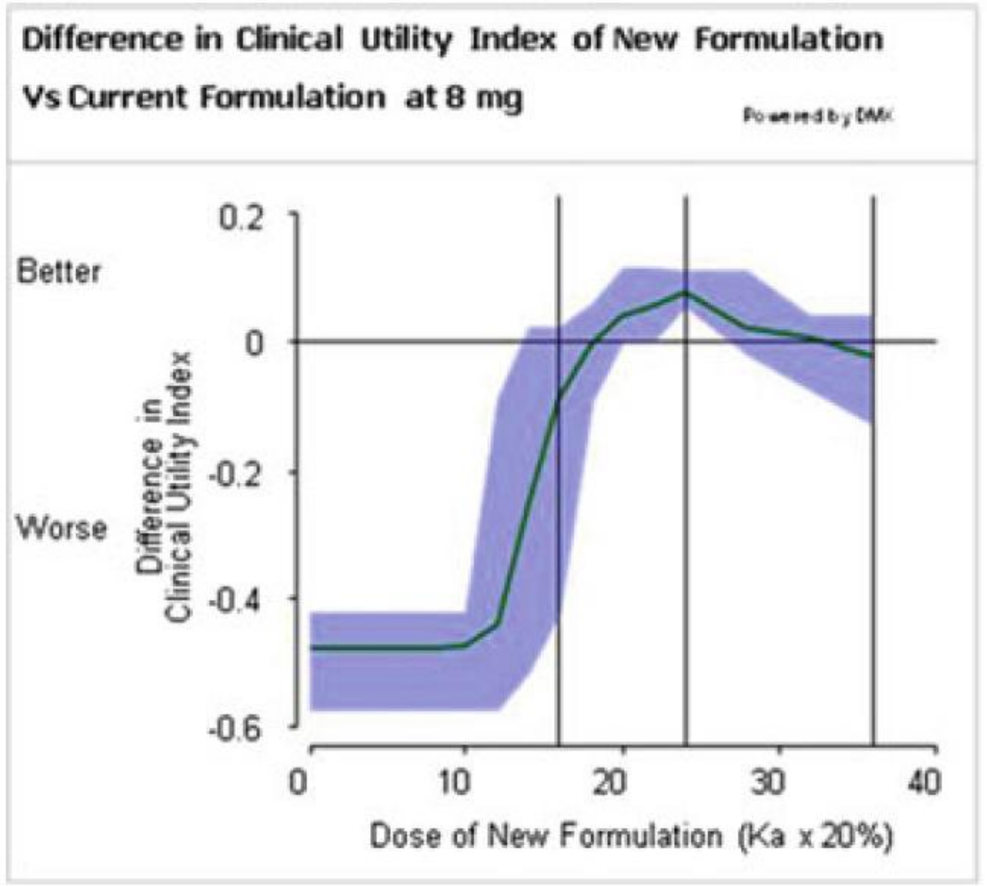
Note: Figure depicts the steps in an integrated CUI-pharmacometric modeling project, with detail on the pharmacometric steps. Literature data is used to guide the construction of the PK and exposure-response models, which are based on subject-level data where possible. “Simulations” incorporates the integration of the CUI model with the PM output, as previously depicted in Figure 5.3 Predicted CUI is the variable of final interest.

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At 20% of the original absorption rate, doses of 17–35 mg beat the current formulation's CUI







Regulators are running the same math

EMA · 2011 decision conference

Belimumab, systemic lupus erythematosus

- **Convened after regulators disagreed.** A one-day MCDA decision conference tested whether a shared model could resolve the benefit–risk impasse.
- **Every effect on one 0–100 scale.** Efficacy and safety measures were each converted to a common preference scale, then explicitly weighted.
- **Quantifying it didn't force an answer.** The benefit–safety scores for the two doses and placebo came out very close — the model clarified the disagreement; it didn't erase it.

FDA · 2020

Ticagrelor (Brilinta), primary prevention

- **First drug decision where FDA used MCDA.** Applied to a supplemental approval for primary prevention of MI or stroke.
- **Six scored outcomes:** CV death, MI, ischemic stroke, fatal bleed, intracranial hemorrhage, other major bleeds — using THEMIS trial data.
- **Explicit weights, explicit sensitivity.** Review-team judgments on relative importance were quantified and stress-tested, not argued qualitatively.
- **Outcome:** approved for primary prevention in high-risk coronary artery disease plus type 2 diabetes patients.

Sources: Lackey LG, Garnett CE, Senatore F. Circulation. 2021;144:655–658 (ticagrelor). Angelis A, et al. Br J Clin Pharmacol. 2021 (belimumab MCDA case study, disguised per publicly available records).

The FDA framework

1. **PDUFA V committed FDA to a structured approach —** the 2013 draft framework[1] formalized five dimensions sponsors and reviewers now walk through together: condition, current options, benefit, risk, and risk management.

2. **It doesn't mandate a method, only a structure —** the framework asks for the dimensions to be laid out explicitly; whether that becomes a qualitative table or a quantitative MCDA/CUI score is the sponsor's choice.

3. **Ticagrelor is the framework in motion —** both MCDA exercises are instances of this structure made quantitative, not a separate regulatory track.

4. **The deck shows the quantitative cases but not the structure that invites them —** naming the framework explains why agencies were already receptive when MCDA/CUI methods arrived.

1 U.S. Food and Drug Administration. Structured Approach to Benefit-Risk Assessment in Drug Regulatory Decision-Making: Draft PDUFA V Implementation Plan (February 2013).

What holds across all examples

1. **Multiple attributes compete. Efficacy**, **safety**, **cost**, **convenience** — no single endpoint settles a formulary, dosing, or approval decision on its own.

2. **Weights are elicited, not assumed.** Experts or regulators state relative importance up front, and sensitivity analysis tests whether the ranking survives disagreement about those weights.

3. **The arithmetic scales.** The same weighted-sum logic runs from a single compound decision (insomnia, antipsychotics) to an agency-level approval (ticagrelor).

4. **Transparency is the deliverable.** The output that matters isn't just the winning option — it's a defensible, auditable account of why.

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Next: personalizing CUI, hands-on

After the break, two hands-on put this same logic to work:

1. **An ADC in oncology — Venky's track, first.** The same weighted-attribute logic, applied to an antibody–drug conjugate.

2. **Personalized CUI — elagolix in endometriosis, next.** Patient-preference weights from a discrete-choice study feed an exposure–response model to pick a personalized dose.

*Same framework, two more domains — see you after **coffee**.*

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